

Article

MaGOS-IDS: A Mahalanobis-Enhanced OpenMax Method for Graph Neural Network-Based Intrusion Detection

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Abstract

Graph Neural Networks achieve strong closed-set accuracy in network intrusion detection but cannot flag zero-day attacks, because the closed-world assumption forces every input into a known class. OpenMax adds an Extreme Value Theory reject option, yet its Euclidean distance ignores the class-conditional covariance that encodes attack-specific structure, which produces unreliable tail models and rejection thresholds. We propose MaGOS-IDS, which extracts topology-aware embeddings with an edge-aware GCN that fuses flow-level edge features directly into message passing, whitens each class with a regularized Mahalanobis distance so the reject decision respects per-class variance and correlation, and calibrates a per-class EVT tail on these distances to set an attack-pattern-aware rejection boundary without a hand-tuned cutoff. We provide a theoretical justification via a peaks-over-threshold argument: whitening removes the per-class covariance dependence of the distance tail, so a single extreme-value tail model calibrates consistently across classes. On three benchmarks (NF-BoT-IoT, CIC-IDS-2017, UNSW-NB15) under withheld zero-day families, MaGOS-IDS raises open-set AU-PR over the Euclidean OpenMax baseline (0.932 vs. 0.848 on UNSW-NB15) while adding negligible inference cost.

Keywords: network security; intrusion detection; graph neural networks; extreme value theory; Mahalanobis distance; open-set recognition; zero-day attack

1. Introduction

The rapid expansion of digital technologies has made the network infrastructure the backbone of modern society, but it also brings increased risks of cyber attacks. Adversaries continuously develop sophisticated intrusions, such as distributed denial-of-service (DDoS) attacks or stealthy data exfiltration that threaten critical services. Intrusion Detection Systems (IDS) [1,2], especially Network IDS (NIDS) [3], are essential to monitor network traffic and identify malicious activity. To be effective against modern threats, an IDS must be automated, intelligent, and able to capture complex attack patterns that evade traditional rule-based methods. These shortcomings have driven the adoption of machine learning (ML) techniques [4], which learn traffic behaviors directly from data, automatically extract discriminative features, and improve detection accuracy [5,6].

Despite their expressive power, IDS remain fundamentally limited by a closed world assumption: they can only recognize attack classes seen during training [7]. A zero-day attack exploits a vulnerability that is absent from the training data. As shown in Figure 1, a



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standard ML-based IDS classifier cannot abstain on such traffic: it must assign the flow to one of the known classes, and in practice it is most often absorbed into the benign class [8].

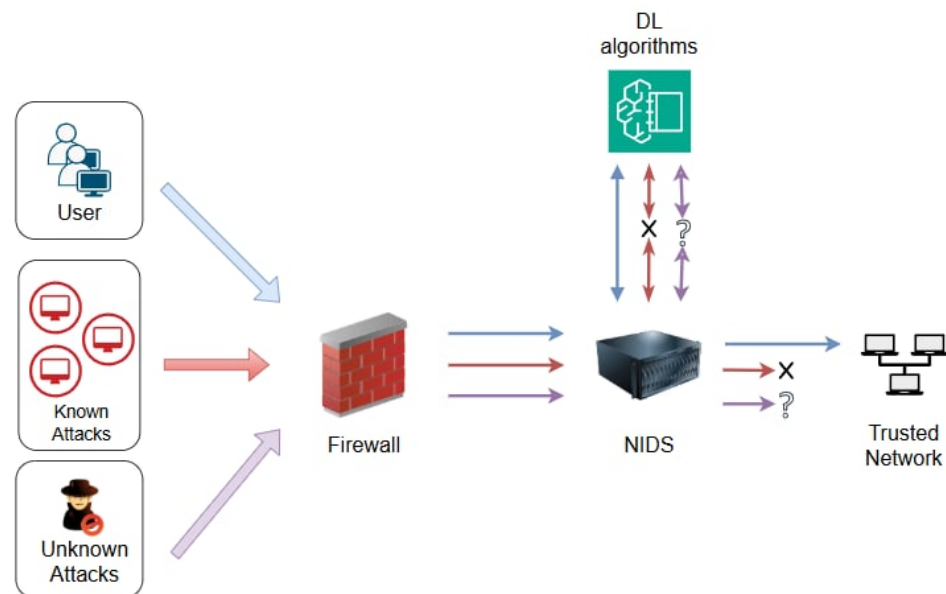


Figure 1. Open-set intrusion detection. Behind the firewall the NIDS sees benign flows (blue), known attack families (red), and zero-day traffic (purple) that belongs to no known class. A closed-set classifier must map every zero-day flow onto a known label; an open-set NIDS flags it as unknown instead. As illustrated, the “X” symbol denotes known attacks that are explicitly blocked, while the “?” symbol represents novel zero-day attacks isolated and flagged for further inspection.

In a security context, such a wrong decision can have severe consequences, allowing undetected intrusions to persist and cause substantial damage. To address this vulnerability, researchers have begun integrating open-set recognition (OSR) capabilities into detection pipelines [9]. By issuing an “unknown” alert rather than forcing an incorrect classification, OSR provides a principled mechanism for flagging potential zero-day attacks, making it a critical component for building robust, real world IDS. OSR introduces two methods, Discriminative methods [10], which establish decision boundaries around known classes and reject samples that fall outside them based on distance or confidence scores, and Generative methods [11], which model the probability distribution of known classes and flag low-likelihood samples as novel. Both paradigms aim to answer the critical question: “is this traffic something I have never seen before?” thus providing a principled mechanism for zero-day detection. Among discriminative OSR techniques, OpenMax [12] has gained particular attention because it grounds the rejection decision in Extreme Value Theory (EVT) [13], using Weibull distribution [14] and reject mechanisms, allowing a statistically rigorous estimation of the probability that a sample originates from an unknown class based on distances in the model’s embedding space.

Historically, learning algorithms, such as Support Vector Machines (SVM) [15,16], Random Forests (RF) [17,18], standard Multi-Layer Perceptrons (MLP) [19–21], Convolutional Neural Network (CNN) [22,23], Recurrent Neural Network (RNN) [24,25], and Long-Short Term Memory (LSTM) [26,27], have been widely deployed within these NIDS frameworks. However, the efficacy of these traditional architectures is severely bottlenecked by their reliance on the independent and identically distributed (i.i.d.) assumption. By evaluating each network flow as an isolated, independent tabular record, conventional models fundamentally fail to capture the complex, relational communication dynamics inherent in sophisticated cyber threats, such as DDoS [28] campaigns, botnet propagation, and lateral movement. Recognizing that modern attacks are structurally distributed rather

than statistically isolated, it has become imperative for defense mechanisms to capture the topological patterns of network traffic. Consequently, Graph Neural Networks (GNNs) have emerged as a paradigm-shifting architecture in the IDS domain [29]. By mapping tabular traffic flows into non-Euclidean graph structures, GNNs leverage spatial message-passing mechanisms to dynamically aggregate feature representations from topological neighborhoods [30]. This profound architectural advantage allows GNN-based IDS [31] to encapsulate both the localized statistical properties of an individual flow and its broader structural context within the network, yielding state-of-the-art detection capabilities for coordinated intrusions.

Discriminative OSR methods, such as OpenMax, typically define class boundaries using distance metrics or confidence scores derived from the model's penultimate layer, rejecting samples that fall too far from any known class prototype. Generative approaches, in contrast, learn the underlying probability distribution of known classes for instance, via autoencoders, generative adversarial networks, or normalizing flows and flag samples with low reconstruction likelihood or high reconstruction error as novel. While both families have been adapted to IDS, their integration with GNN-based architectures remains shallow and faces several critical shortcomings. We identify the following four limitations:

- Neglect of class-conditional covariance structure: Existing distance-based OSR methods, including standard OpenMax, rely on Euclidean distance, which implicitly assumes spherical, uncorrelated embeddings. This ignores the natural variance and correlations induced by different attack patterns, leading to poor tail modeling and unreliable unknown probability estimates.
- Inconsistent Weibull tail fitting across classes: The distribution of Euclidean distances within a class depends on the eigenvalue spectrum of that class's covariance matrix. As a result, the shape of the distance tail can vary substantially from one attack class to another, making a single Weibull distribution a poor fit for many classes and undermining the theoretical foundation of OpenMax.
- Heuristic and attack-agnostic rejection threshold: In current OpenMax pipelines, the final rejection threshold is typically chosen through grid search or a fixed probability cutoff. This approach lacks a formal connection to the underlying attack patterns or to the statistical behavior of the embedding space, limiting both the interpretability and the adaptability of the rejection mechanism.
- Underutilization of edge-rich embeddings: Even when GNN-based IDS incorporate rich flow-level edge features such as byte counts, packet rates, and protocol flags the subsequent Euclidean distance treats all resulting embedding dimensions as equally important. This fails to exploit the fine-grained variance structure that these features create, effectively discarding the very attack-specific footprint that the GNN was designed to capture.

Recognizing the above challenges, this paper proposes MaGOS-IDS, a Mahalanobis Graph Open-Set IDS that integrates an edge-aware graph convolutional network (GCN) with a Mahalanobis-enhanced OpenMax module. The architecture first constructs flow-level graphs that retain rich edge features such as duration, bytes, packets, and protocol flags and processes them through a GCN that incorporates these features directly into the message-passing step. This produces embeddings that faithfully encode both the topological structure and the fine-grained communication patterns of network traffic. At the open-set stage, MaGOS-IDS replaces the standard Euclidean distance with a regularized Mahalanobis distance [32], normalizing the embedding space according to each class's covariance structure. This enables a more faithful Weibull tail model and, consequently, a principled, attack-pattern-aware rejection threshold grounded in EVT.

To the best of our knowledge, MaGOS-IDS is the first framework to jointly combine graph-based topological embeddings from a GNN with Mahalanobis distance-based OSR, while providing a formal theoretical justification linking the two. Prior work such as OIDMD [33], OpenIDS [34], CLUSTER [35], InfoGAN [36], DQN-IDS [37], CLOSR [38] have applied Mahalanobis distance within intrusion detection pipelines; however, these methods operate on flat, tabular feature representations and do not leverage the relational structure of network traffic. Critically, no existing work has established a principled foundation for the rejection threshold in a GNN-based IDS setting. We do not assume the squared Mahalanobis distances over GNN embeddings are exactly χ^2 -distributed; instead we treat the Gaussian model as a reference and derive the tail model from a peaks-over-threshold argument (Section 3.3), under which per-class whitening removes the dependence of the exceedance tail on each class's covariance spectrum. This yields a rejection mechanism that is simultaneously topology-aware and covariance-calibrated, a combination absent from prior intrusion detection literature.

We position our contribution precisely against prior art. The edge-aware GCN feature extractor (Section 4.2) adapts established edge-featured message passing [39], and the OpenMax reject mechanism itself originates with [12]; Mahalanobis distance for open-set intrusion detection has also been used on tabular features by OIDMD. The genuine methodological advance of MaGOS-IDS is the coupling of these elements: applying per-class Mahalanobis-whitened distances to graph-level GNN embeddings, and grounding the reject threshold in a peaks-over-threshold EVT argument tailored to the resulting distance distribution. We do not claim the individual components as novel; we claim their integration and its theoretical justification in the GNN-based IDS setting, which no prior work provides.

The primary contributions of this research address the four limitations enumerated above in one-to-one correspondence, and are summarized as follows:

- We fused flow-level edge features (duration, bytes, packets, protocol, TCP flags) directly into the GCN message-passing step, so the resulting per-class embedding covariance retains the attack-specific footprint that Euclidean OpenMax discards.
- We replaced the Euclidean distance in the open-set stage by a regularized per-class Mahalanobis distance, whitening the embedding space so that the reject decision accounts for per-class variance and correlation rather than assuming spherical classes.
- We fitted EVT tail per class to the Mahalanobis-whitened distances, making the tail model comparable across classes instead of fluctuating with each class's covariance spectrum.
- We also aggregated rejected logit mass into an unknown pseudo-logit that competes directly in the softmax argmax, which removes the heuristic cutoff and lets the rejection boundary adapt per sample.

We validate each design choice through an ablation study and EVT goodness-of-fit tests, and we benchmark against twelve GNN and OSR baselines across three datasets.

The remainder of this manuscript is organized as follows: Section 2 reviews the existing literature on graph-based IDS and OSR. Section 3 establishes the fundamental theoretical preliminaries underpinning this research. Section 4 details the mathematical formulation and architectural pipeline of the proposed MaGOS-IDS framework. Section 5 outlines the experimental design, including dataset configurations, baseline models, and the strict open-set evaluation protocol. Section 6 provides a comprehensive performance evaluation and discusses the empirical results of the proposed zero-day rejection mechanism. Finally, Section 7 summarizes the contributions of this study and outlines trajectories for future work.

2. Related Work

2.1. GNN-Based Network Intrusion Detection

Table 1 positions MaGOS-IDS against representative prior work along three axes that jointly determine open-set behaviour: whether the input is modelled relationally or as flat records, whether class-conditional second-order structure enters the reject decision, and how the reject threshold is obtained. The graph-based methods in Category 1 exploit topology but retain a closed-world softmax, so they provide no reject option at all. The open-set methods in Category 2 provide a reject option but operate on tabular features and therefore discard topology; of these, only OIDMD models class-specific covariance, and it does so without an EVT-calibrated tail. MaGOS-IDS is the only entry that occupies all three cells simultaneously.

Table 1. Comparison of representative intrusion detection methods. Category 1 methods are all graph-based with no zero-day capability; Category 2 methods are all tabular with a zero-day reject option. MaGOS-IDS is the only method combining a graph-topological representation, class-specific covariance modelling, and an EVT-based reject option.

Method	Cov. Model	Reject Mechanism	Key Limitation
<i>Category 1: Graph-based closed-set models (graph repr., no zero-day capability)</i>			
EE-GCN [39]	None	Softmax	No reject option
E-ResSAGE [40]	None	Softmax	No reject option
E-ResGAT [40]	None	Softmax	No reject option
FN-GNN [41]	None	Softmax	No reject option
NIGNN [42]	None	Softmax	Ignores edge features
CAGN-GAT [43]	None	Softmax/Dist.	Uncalibrated distance
<i>Category 2: Tabular open-set recognition models (tabular repr., zero-day capable)</i>			
OpenIDS [34]	None	Pseudo-EVT	Discards topology
CLUSTER [35]	None	Centroid Dist.	Spherical boundaries
OIDMD [33]	Class-spec.	Mahalanobis	No tail calibration
InfoGAN+OpenMax [36]	None	Euclidean OpenMax	Spherical boundaries
DQN-IDS [37]	None	RL Policy	No statistical guarantee
CLOSR [38]	None	Orthogonal	Hand-tuned cutoff
MaGOS-IDS (Proposed)	Class-spec.	Mahalanobis + EVT	Snapshot-level labels

Traditional machine learning approaches to network intrusion detection have historically evaluated network flows as isolated, tabular records. While computationally inexpensive, this methodology inherently ignores the spatial and structural dependencies between communicating endpoints. To address this fundamental limitation, recent literature has increasingly adopted GNN to map the complex topological patterns of modern cyberattacks, transforming discrete NetFlow logs into interconnected spatial structures where IP addresses serve as nodes and communication flows serve as edges.

Foundational frameworks in this domain, such as the Network Intrusion Graph Neural Network (NIGNN) [42], demonstrated that adopting this topological perspective significantly improves the detection of distributed attacks (e.g., botnets, coordinated DDoS). By propagating spatial messages across the network topology, NIGNN successfully captured the global receptive field of the network, rather than relying on localized packet inspection. However, researchers quickly recognized that relying solely on simple adjacency matrices omitted critical flow metadata. To resolve this, subsequent architectures evolved to incorporate multi-dimensional edge characteristics. For instance, the Edge-Featured Graph Convolutional Network (EE-GCN) [39] natively integrated flow-level statistics such as total byte counts, packet duration, and inter-arrival times directly into the spatial convolution phase. This advancement allowed the model to yield highly contextualized latent represen-

tations that captured both the structure of the connection and the volumetric behavior of the payload.

As GNN architectures grew deeper to capture multi-hop attack paths, they encountered the well-documented issue of “oversmoothing,” where node representations become increasingly indistinguishable. To prevent this degradation, researchers integrated residual connections into advanced spatial aggregators. The E-ResSAGE architecture [40] combined edge-feature integration with GraphSAGE neighborhood sampling and residual pathways, ensuring highly efficient and scalable anomaly detection across massive, resource-constrained IoT networks. Similarly, E-ResGAT [40] utilized graph attention mechanisms alongside these residual connections. This allowed the network to dynamically weigh the importance of specific network connections, selectively focusing on malicious routing paths while suppressing the noise from benign background traffic during feature aggregation.

Recent advancements have focused on optimizing this representation space even further to maximize multi-class separation. The Flow-Node Graph Neural Network (FN-GNN) [41,44] explicitly models the dual relationship between node-level host statistics and edge-level flow paths to maximize feature co-extraction, treating flows and hosts as mutually informative entities. Furthermore, highly complex hybrid models like CAGN-GAT [43] have fused contrastive learning protocols with attention mechanisms. By strategically pulling samples of the same attack family together and pushing different known families apart in the latent space, CAGN-GAT achieves exceptional multi-class classification accuracy and minimizes intra-class variance.

2.2. Open-Set Recognition in Network Intrusion Detection

As the structural vulnerabilities of the closed world assumption become increasingly apparent, the cybersecurity community has recently begun integrating OSR paradigms into intrusion detection architectures. The primary objective of these systems is to maintain high classification accuracy on known threats while establishing robust rejection boundaries to isolate zero-day anomalies.

Early adaptations in this domain primarily relied on augmenting standard deep learning features with traditional open-set algorithms. For instance, researchers have proposed combining unsupervised generative models, such as InfoGAN, with the standard OpenMax algorithm to calculate the probability of unknown traffic based on reconstructed tabular features [36]. Recognizing the limitations of standard Softmax distributions, subsequent literature shifted towards explicitly modeling latent class distributions. For example, the OIDMD method establishes a rejection mechanism that isolates unknown intrusions by measuring the Mahalanobis distance against established known class covariances within flat, tabular feature spaces [33]. Similarly, recent advancements in intelligent vehicle security have formalized open-world intrusion detection on CAN buses by utilizing the distance from inputs to cluster centroids as a primary training constraint [35,45].

To further refine these decision boundaries, state-of-the-art research has explored advanced optimization techniques, including EVT and contrastive learning. The OpenIDS framework utilized a MinMax Autoencoder coupled with a Pseudo Extreme Value Machine to establish statistical rejection thresholds for network anomalies [34]. Meanwhile, the recently proposed CLOSAR architecture leverages contrastive learning to force known classes into distinct embedded subspaces, theoretically causing zero-day payloads to manifest as orthogonal vectors [38]. Other novel approaches have even bypassed thresholding entirely, deploying Deep Q-Networks (DQN) to dynamically learn separation policies for unknown traffic via reinforcement learning [37].

3. Preliminaries

3.1. Problem Formulation

We model network traffic as a sequence of graph snapshots. For a fixed time window Δt , a directed graph $\mathcal{G} = (V, E, \mathbf{X}_v, \mathbf{X}_e)$ is constructed, where V is the set of unique IP addresses observed during Δt , and $E \subseteq V \times V$ contains a directed edge (i, j) for every communication flow from IP i to IP j . The matrix $\mathbf{X}_v \in \mathbb{R}^{|V| \times d_v}$ provides initial node features, such as aggregated statistics of flows terminating at or originating from each IP, while $\mathbf{X}_e \in \mathbb{R}^{|E| \times d_e}$ carries edge features describing each flow (e.g., duration, byte count, packet count, transport protocol, and TCP flags). Let $\mathcal{K} = \{1, 2, \dots, C\}$ denote the set of known classes, comprising normal traffic and a collection of known attack types. During training, every graph $\mathcal{G}_i$ is assigned a label $y_i \in \mathcal{K}$. At test time, the system may encounter graphs from classes in $\mathcal{K}$ or from unknown (zero-day) attack types, formally denoted $y = \text{unknown} \notin \mathcal{K}$.

The objective of the proposed framework is to learn a mapping function $f : \mathcal{G} \rightarrow \mathcal{K} \cup \{\text{unknown}\}$. To achieve this, the model must jointly minimize two distinct risk factors. Formally, the optimal function f^* is derived by solving:

$$f^* = \arg \min_f [\mathcal{R}_{\mathcal{E}}(f) + \gamma \mathcal{R}_{\mathcal{O}}(f)]. \quad (1)$$

where γ is a regularization hyperparameter balancing the two objectives.

1. Empirical Risk ($\mathcal{R}_{\mathcal{E}}$): This term represents the closed-set classification error on the known training data. It ensures the model accurately classifies established network topologies. Given a labeled training set of N graphs $\mathcal{D}_{\text{train}} = \{(\mathcal{G}_i, y_i)\}_{i=1}^N$ where $y_i \in \mathcal{K}$, the empirical risk is defined using the standard cross-entropy loss $\mathcal{L}$:

$$\mathcal{R}_{\mathcal{E}}(f) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f(\mathcal{G}_i), y_i). \quad (2)$$

where $\mathcal{L}(\cdot, \cdot)$ denotes the categorical cross-entropy loss between the model's predicted class distribution and the ground-truth label:

$$\mathcal{L}(f(\mathcal{G}_i), y_i) = - \sum_{c \in \mathcal{K}} \mathbf{1}[y_i = c] \log P(y = c \mid \mathcal{G}_i), \quad (3)$$

with $\mathbf{1}[\cdot]$ the indicator function and $P(y = c \mid \mathcal{G}_i)$ the softmax probability assigned to class c for input graph $\mathcal{G}_i$.

2. Open Space Risk ($\mathcal{R}_{\mathcal{O}}$): This term quantifies the probability of misclassifying an unknown, zero-day attack as one of the known classes in $\mathcal{K}$. Let $\mathcal{S}$ denote the finite topological space occupied by known training samples, and $\mathcal{O}$ denote the infinite open space of unknown topologies (where $\mathcal{S} \cap \mathcal{O} = \emptyset$). Following the foundational definition of OSR, the open space risk is the relative measure of the positively labeled area that extends into the unknown space:

$$\mathcal{R}_{\mathcal{O}}(f) = \frac{\int_{\mathcal{O}} f(\mathcal{G}) d\mathcal{G}}{\int_{\mathcal{S} \cup \mathcal{O}} f(\mathcal{G}) d\mathcal{G}}. \quad (4)$$

Here $\mathcal{S}$ is the compact support region occupied by correctly classified training graphs in the latent space, $\mathcal{O}$ is the complementary open space of unobserved topologies (with $\mathcal{S} \cap \mathcal{O} = \emptyset$), and $f(\mathcal{G}) \in \{0, 1\}$ indicates whether the model positively labels $\mathcal{G}$ as a known class. The numerator therefore measures the volume of open space incorrectly claimed by the classifier, which a standard softmax maximizes by extending its decision boundaries without bound.

Because computing an integral over the infinite open space $\mathcal{O}$ is mathematically intractable in high-dimensional graph latent spaces, a standard GNN classifier inherently ignores $\mathcal{R}_{\mathcal{O}}(f)$, causing the decision boundaries to expand infinitely. Consequently, the proposed MaGOS-IDS architecture explicitly bounds the known support space $\mathcal{S}$ via EVT, effectively minimizing the numerator $\int_{\mathcal{O}} f(\mathcal{G}) d\mathcal{G}$ without degrading the empirical risk.

3.2. Standard Euclidean OpenMax

OpenMax equips a deep network with a post-processing layer that estimates how likely a test sample is to belong to an unknown class. Assume we have a trained encoder $g_{\theta}: \mathcal{G} \rightarrow \mathbf{v}$ that maps an input graph to a penultimate embedding vector $\mathbf{v} \in \mathbb{R}^k$. The final classification layer is a standard softmax trained on the known classes $\mathcal{K}$.

For each known class $c \in \mathcal{K}$, the mean activation vector (MAV) is computed from the correctly classified training samples:

$$\boldsymbol{\mu}_c = \frac{1}{N_c} \sum_{i: y_i=c} \mathbf{v}_i. \quad (5)$$

The original OpenMax then measures the Euclidean distance between a test embedding $\mathbf{v}$ and each class prototype:

$$d_E(\mathbf{v}, c) = \|\mathbf{v} - \boldsymbol{\mu}_c\|_2. \quad (6)$$

EVT is used to model the tail of these distances. For class c , the η largest distances among correctly classified training samples are collected, and a Weibull distribution $\text{Weibull}(\kappa_c, \lambda_c)$ is fitted to them using maximum likelihood. The Weibull cumulative distribution function (CDF) with $d \geq 0$ is

$$F(d; \kappa, \lambda) = 1 - \exp(-(d/\lambda)^\kappa). \quad (7)$$

At test time, the softmax layer produces a preliminary best known class $c^* = \arg \max_{c \in \mathcal{K}} P(y = c | \mathbf{v})$. The probability that the sample is unknown is then estimated as

$$P_{\text{unknown}} = F(d_E(\mathbf{v}, c^*); \kappa_{c^*}, \lambda_{c^*}). \quad (8)$$

If P_{unknown} exceeds a pre-defined threshold τ , the sample is rejected as unknown; otherwise it is assigned label c^* .

A fundamental limitation of this standard formulation is that the Euclidean distance implicitly treats all feature dimensions as uncorrelated and equally scaled, ignoring the class-conditional covariance structure that different attack patterns induce in the embedding space. This observation directly motivates replacing Euclidean distance with the Mahalanobis distance, which explicitly accounts for second-order information.

3.3. Mahalanobis Distance and Its Statistical Advantage

For a class c with mean $\boldsymbol{\mu}_c$ and a regularised covariance matrix $\boldsymbol{\Sigma}_c$ estimated from the embeddings of its training samples, the Mahalanobis distance of a test embedding $\mathbf{v}$ to class c is defined as

$$d_M(\mathbf{v}, c) = \sqrt{(\mathbf{v} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\mathbf{v} - \boldsymbol{\mu}_c)}. \quad (9)$$

Intuitively, this metric first applies a whitening transformation $\boldsymbol{\Sigma}_c^{-1/2}(\mathbf{v} - \boldsymbol{\mu}_c)$ that decorrelates the features and scales them to unit variance, then computes the ordinary Euclidean distance in the whitened space.

The statistical advantage of the Mahalanobis distance follows from a mild assumption about the penultimate embeddings. If the class-conditional embeddings are approximately

multivariate Gaussian, $\mathbf{v} \mid c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$, then the squared Mahalanobis distance is approximately chi-squared with k degrees of freedom,

$$d_M^2(\mathbf{v}, c) \approx \chi_k^2, \quad (10)$$

where k is the embedding dimension. We stress that this is a reference model, not an exact law: real GNN embeddings depart from Gaussianity, most visibly in the tail, which is precisely the region that governs the reject decision. Rather than trusting the closed-form χ_k^2 quantile, which would be valid only under exact Gaussianity, we calibrate the tail empirically. This is the role of Extreme Value Theory: it provides a distributional model for the exceedances that is robust to moderate departures from the Gaussian reference.

Concretely, we adopt a peaks-over-threshold (POT) formulation. By the Pickands–Balkema–de Haan theorem [46], for a broad class of parent distributions the exceedances of a random variable over a high threshold u converge, as u grows, to a Generalized Pareto Distribution (GPD),

$$\Pr(d_M - u \leq x \mid d_M > u) \longrightarrow G_{\zeta, \sigma}(x) = 1 - \left(1 + \frac{\zeta x}{\sigma}\right)^{-1/\zeta}, \quad (11)$$

with shape ζ and scale σ . For a light-tailed parent such as the chi distribution, the shape parameter satisfies $\zeta \rightarrow 0$, which corresponds to an exponential-type (Gumbel-domain) tail decay. In this regime the exceedance distribution is equivalently well described by a two-parameter Weibull CDF fitted to the upper-tail distances, which we retain as the working model for consistency with the OpenMax formulation. The advantage of the Mahalanobis distance is that whitening removes the dependence of the tail shape on the per-class covariance spectrum: all classes are mapped to a common, approximately isotropic geometry before the tail is fitted, so a single family of tail models applies consistently across classes. We therefore expect the fitted Weibull shape parameters to be stable and concentrated across classes, rather than fluctuating with each class's covariance as they do under the Euclidean metric.

By contrast, the Euclidean distance squared $d_E^2 = \|\mathbf{v} - \boldsymbol{\mu}_c\|^2$ is a generalized chi-squared statistic whose distribution depends entirely on the unpredictable eigenvalue spectrum of $\boldsymbol{\Sigma}_c$. Different attack types induce highly skewed and distinct covariance structures; for instance, a DDoS attack exhibits high variance in flow-volume dimensions, whereas a port scan varies primarily along port-access dimensions. As a result, the tail shape of the Euclidean distances fluctuates arbitrarily across classes, causing a universal Weibull distribution to fit poorly, which undermines the theoretical foundation of the OpenMax probability estimates.

As illustrated in Figure 2, the dashed circles represent rigid Euclidean rejection boundaries, while the solid ellipses show high Mahalanobis confidence regions that dynamically adapt to each class's distinct covariance matrix. A never-seen sample (red point) that falls outside the Euclidean boundary of the blue class is correctly rejected by its Mahalanobis ellipse, demonstrating how the Mahalanobis distance yields a class-invariant, theoretically well-behaved distance tail that strictly adheres to the mathematical prerequisites of EVT.

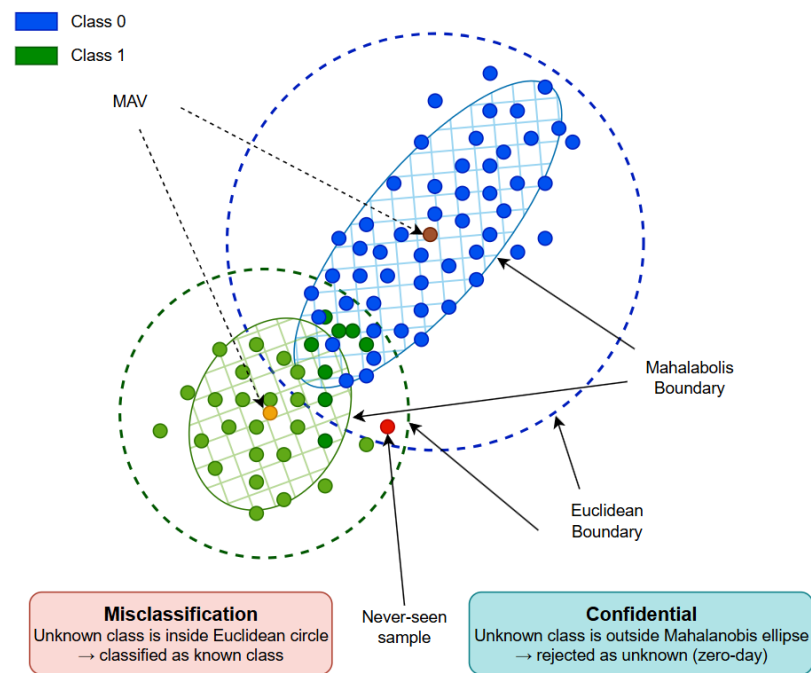


Figure 2. Euclidean vs. Mahalanobis contours for open-set rejection.

4. Proposed Method

4.1. Model Architecture

As illustrated in Figure 3, the process begins by transforming raw network data into an attributed topological graph where IP addresses serve as nodes and communication flows act as connecting edges. During training, a GCN layer with the ReLU activation function dynamically aggregates these edge features to extract deep, structurally aware latent node embeddings. Rather than relying solely on a standard closed-set softmax classifier, MaGOS-IDS extracts the activation vectors to calculate the MAV for each known traffic class. To accurately capture the class-conditional covariance of the network features, the system computes the Mahalanobis distance between training samples and their respective MAVs, subsequently applying EVT to fit a Weibull distribution to the extreme tail distances. During testing, as new topological embeddings are extracted from dynamic traffic, the model evaluates them against these calibrated Weibull distributions; nodes falling within acceptable probability bounds are classified into known categories, while those located in the extreme statistical tail are explicitly rejected and isolated as unknown zero-day attacks.

The upper path is training and the lower path is inference. Node colours follow the legend: green is the victim host, red a known attacker, white a normal host, blue an unknown (zero-day) attacker, and peach a host assigned to a known class. Lavender nodes inside the GCN blocks are intermediate node states during message passing and carry no class meaning. Coloured bars are per-node feature vectors; in the MAV extraction block each row is the activation vector of one known class and the highlighted bar is the resulting mean activation vector. Solid and dashed curves in the distribution panels are the fitted per-class Weibull tails. In the output panels, “Classify Learned class” shows a snapshot whose nodes all fall inside the calibrated boundaries, whereas “Classify Unknown class” shows a snapshot containing one node in the extreme tail, which is rejected as a zero-day.

4.2. Edge-Aware Graph Convolutional Feature Extraction

While traditional GCN effectively capture topological structures, they primarily aggregate node features and treat edges merely as binary adjacency links. As established in our

problem formulation, the critical intrusion signatures in network traffic, such as duration, byte counts, and TCP flags, reside within the flow-based edge matrix X_e . To preserve this essential diagnostic information, MaGOS-IDS utilizes an Edge-Aware Graph Convolutional architecture that dynamically fuses both host-level and flow-level attributes during the spatial message-passing phase.

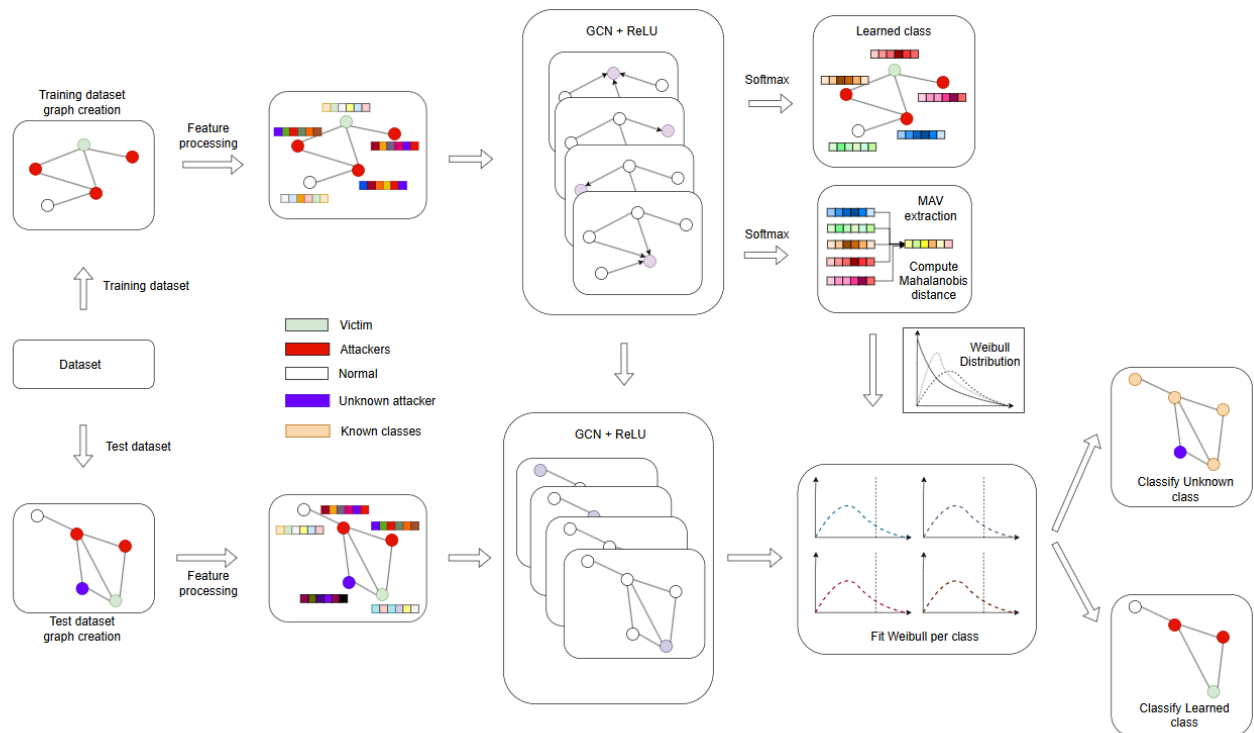


Figure 3. MaGOS-IDS architecture.

Let $\mathbf{h}_i^{(l)}$ denote the latent feature representation of node $i \in V$ at the l -th convolutional layer. At the initial layer ($l = 0$), the node representations are initialized using the raw vertex features, $\mathbf{h}_i^{(0)} = \mathbf{x}_{v,i} \in \mathbf{X}_v$.

During each forward propagation step, the network constructs a highly contextualized message $\mathbf{m}_{ji}^{(l)}$ sent from every neighboring node $j \in \mathcal{N}(i)$ to the target node i . This is achieved by concatenating the neighbor's hidden state with the static flow features of the connecting edge $\mathbf{e}_{ji} = \mathbf{x}_{e,(j,i)} \in \mathbf{X}_e$, followed by a learnable linear transformation:

$$\mathbf{m}_{ji}^{(l)} = \mathbf{W}_{\text{msg}}^{(l)} [\mathbf{h}_j^{(l-1)} \parallel \mathbf{e}_{ji}], \quad (12)$$

where $\parallel$ denotes the concatenation operator and $\mathbf{W}_{\text{msg}}^{(l)}$ is the weight matrix specific to the edge-fusion operation at layer l .

Subsequently, the network aggregates these incoming messages from the immediate topological neighborhood $\mathcal{N}(i)$. To prevent the structural distortion caused by exploding gradients on high-degree nodes (a common topological signature of DDoS attacks and port scans), we employ a mean-pooling aggregator:

$$\mathbf{a}_i^{(l)} = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ji}^{(l)}. \quad (13)$$

The target node updates its own latent representation by concatenating its previous state with the aggregated neighborhood message, applying a layer-specific weight ma-

trix $\mathbf{W}_{\text{update}}^{(l)}$ and passing the result through a Rectified Linear Unit (ReLU) non-linear activation function:

$$\mathbf{h}_i^{(l)} = \text{ReLU}\left(\mathbf{W}_{\text{update}}^{(l)} \left[\mathbf{h}_i^{(l-1)} \parallel \mathbf{a}_i^{(l)} \right]\right). \quad (14)$$

After L layers of spatial message passing, the network yields a deep, structurally aware embedding $\mathbf{h}_i^{(L)}$ for every IP address in the time window Δt . Because our fundamental objective requires classifying the entire graph snapshot $\mathcal{G}$, we apply a global readout (pooling) function to aggregate the node-level embeddings into a unified graph-level representation vector $\mathbf{z}_{\mathcal{G}} \in \mathbb{R}^{d_z}$:

$$\mathbf{z}_{\mathcal{G}} = \text{READOUT}\left(\left\{\mathbf{h}_i^{(L)} \mid i \in V\right\}\right) = \frac{1}{|V|} \sum_{i \in V} \mathbf{h}_i^{(L)}. \quad (15)$$

This final latent vector $\mathbf{z}_{\mathcal{G}}$ seamlessly encapsulates both the statistical flow properties and the spatial routing topology of the network window. It serves as the direct input into the Mahalanobis covariance calibration module to isolate zero-day attack footprints.

Algorithm 1 details the training procedure for the Edge-Aware GCN. The network initializes node representations from the raw host features (Step 1) and iteratively updates them through spatial message passing (Step 2). Crucially, during message construction, the network dynamically fuses the hidden state of each neighbor with its corresponding edge features (the flow statistics), ensuring that critical intrusion signatures are preserved. After L convolutional layers, a global readout function (Step 3) aggregates the highly contextualized node embeddings into a singular, graph-level activation vector $\mathbf{z}_{\mathcal{G}}$. This final vector represents the complete topological and statistical state of the network window Δt , serving as the direct input for classification and subsequent Mahalanobis covariance calibration.

Algorithm 1 Edge-aware GCN training and activation vector extraction.

Require: Dataset $\mathcal{D}_{\text{train}} = \{(G_k, y_k)\}_{k=1}^K$, $y_k \in \mathcal{K}$; layers L ; learning rate α ; epochs M .

Ensure: Trained weights $\Theta = \{\mathbf{W}_{\text{msg}}^{(l)}, \mathbf{W}_{\text{update}}^{(l)}\}_{l=1}^L$ and classification head.

```

1: for epoch = 1 to M do
2:   for each graph  $\mathcal{G}_k \in \mathcal{D}_{\text{train}}$  do
3:     Step 1: Feature Initialization
4:      $\mathbf{h}_i^{(0)} \leftarrow \mathbf{x}_{v,i}$  for all  $i \in V$ 
5:     Step 2: Spatial Message Passing
6:     for  $l = 1$  to  $L$  do
7:       for each node  $i \in V$  do
8:          $\mathbf{m}_{ji}^{(l)} \leftarrow \mathbf{W}_{\text{msg}}^{(l)} [\mathbf{h}_j^{(l-1)} \parallel \mathbf{e}_{ji}]$  ▷ Equation (12)
9:          $\mathbf{a}_i^{(l)} \leftarrow \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ji}^{(l)}$ 
10:         $\mathbf{h}_i^{(l)} \leftarrow \text{ReLU}(\mathbf{W}_{\text{update}}^{(l)} [\mathbf{h}_i^{(l-1)} \parallel \mathbf{a}_i^{(l)}])$  ▷ Equation (14)
11:      end for
12:    end for
13:    Step 3: Graph-Level Readout
14:     $\mathbf{z}_{\mathcal{G}_k} \leftarrow \frac{1}{|V|} \sum_{i \in V} \mathbf{h}_i^{(L)}$  ▷ consumed by Mahalanobis calibration
15:    Step 4: Classification and Loss
16:     $\hat{y}_k \leftarrow \text{Softmax}(\mathbf{W}_{\text{cls}} \mathbf{z}_{\mathcal{G}_k} + \mathbf{b}_{\text{cls}})$ 
17:     $\mathcal{L}_k \leftarrow -\sum_{c \in \mathcal{K}} y_{k,c} \log \hat{y}_{k,c}$ 
18:  end for
19:  Step 5: Parameter Update
20:   $\mathcal{L}_{\text{total}} \leftarrow \frac{1}{K} \sum_{k=1}^K \mathcal{L}_k$ 
21:   $\Theta \leftarrow \Theta - \alpha \nabla_{\Theta} \mathcal{L}_{\text{total}}$ 
22: end for
23: return  $\Theta$ 

```

Following the extraction of the graph-level activation vector $\mathbf{z}_{\mathcal{G}} \in \mathbb{R}^{d_z}$ from the Edge-Aware GCN, standard closed-set architectures project this vector through a fully connected layer to yield a logit vector $\mathbf{v} = [v_1, v_2, \dots, v_C]$, which is subsequently normalized by a Softmax function. Because Softmax enforces a strictly bounded probability distribution

$\sum_{c \in \mathcal{K}} P(y = c | \mathcal{G}) = 1$, it inherently denies the existence of an open space, misclassifying zero-day attacks with artificially high confidence.

To systematically bound this open space risk, MaGOS-IDS intercepts the latent representations *before* softmax normalization to construct geometric boundaries around the known classes. Let $S_c = \{\mathbf{z}_{\mathcal{G},k} \mid y_k = c \text{ and } \hat{y}_k = c\}$ denote the set of activation vectors for all training graphs correctly classified as class $c \in \mathcal{K}$. We first define the geometric centroid of this class by calculating its MAV, denoted as μ_c :

$$\mu_c = \frac{1}{|S_c|} \sum_{\mathbf{z} \in S_c} \mathbf{z}. \quad (16)$$

4.3. Mahalanobis-Based Open Space Calibration

Traditional OpenMax formulations calculate the distance between a new sample and μ_c using the Euclidean distance metric. However, network traffic features such as byte counts and flow durations naturally exhibit distinct, class-conditional variances and are often highly correlated. Euclidean distance imposes a rigid, spherical assumption on the latent space, failing to capture these asymmetric distributions. To address this structural flaw, MaGOS-IDS leverages the Mahalanobis distance, which scales the spatial distance by the inverse of the class covariance matrix Σ_c .

To ensure numerical stability and avoid singular covariance matrices, a common issue in high-dimensional latent spaces, we apply Tikhonov regularization to the covariance matrix by adding a small identity matrix ϵI :

$$\tilde{\Sigma}_c = \frac{1}{|S_c| - 1} \sum_{\mathbf{z} \in S_c} (\mathbf{z} - \mu_c)(\mathbf{z} - \mu_c)^\top + \epsilon I, \quad (17)$$

where $\epsilon > 0$ is a small regularization constant. The Mahalanobis distance d_M for any latent vector $\mathbf{z}$ to class c is therefore formulated as

$$d_M(\mathbf{z}, \mu_c) = \sqrt{(\mathbf{z} - \mu_c)^\top \tilde{\Sigma}_c^{-1} (\mathbf{z} - \mu_c)}. \quad (18)$$

The utilization of the Mahalanobis distance provides a critical theoretical advantage for EVT. Under the approximate Gaussian reference of Section 3.3, the squared Mahalanobis distance is approximately χ^2 -distributed. As argued there, we do not rely on the exact χ^2 quantile; instead we fit the upper tail of the observed distances with a two-parameter Weibull model, consistent with the peaks-over-threshold justification. Because whitening removes the per-class covariance dependence of the tail shape, this single tail family applies consistently across classes, yielding tighter and more stable open-space boundaries than the Euclidean metric, whose tail shape varies with each class's eigenvalue spectrum.

To model this tail, we compute d_M for all $\mathbf{z} \in S_c$ and sort them in descending order. We extract the top- η largest distances (the extreme tail) to form the calibration set $\mathcal{T}_c$. Using Maximum Likelihood Estimation (MLE), we fit these extreme values to a Weibull probability density function, yielding the class-specific scale parameter λ_c and shape parameter κ_c .

During online inference, when a new graph snapshot $\mathcal{G}_{\text{test}}$ is processed, the model computes its Mahalanobis distance to the known MAVs and evaluates it against the calibrated Weibull CDF:

$$\omega_c(\mathbf{z}) = 1 - \exp\left(-\left(\frac{d_M(\mathbf{z}, \mu_c)}{\lambda_c}\right)^{\kappa_c}\right). \quad (19)$$

Here, $\omega_c(\mathbf{z}) \in [0, 1]$ represents the probability that the sample is an anomaly with respect to class c . Following the OpenMax protocol, we recalibrate the top- α original logit

scores by penalizing them based on this Weibull probability. The rejected logit mass is aggregated to formulate a distinct pseudo-activation score for the unknown class:

$$\hat{v}_{\text{unknown}} = \sum_{i=1}^{\alpha} v_{(i)} \cdot \omega_{(i)}(\mathbf{z}). \quad (20)$$

where $v_{(i)}$ represents the i -th largest original logit. Finally, a standard Softmax function is applied over the expanded label space $\mathcal{K} \cup \{\text{unknown}\}$. If $\hat{v}_{\text{unknown}}$ yields the maximum probability, the graph snapshot is definitively rejected as a zero-day intrusion. Crucially, this formulation eliminates the need for an arbitrary, heuristic rejection threshold (τ). Instead, the argmax operator establishes a *dynamic, principled threshold* that directly balances the model's closed-set confidence against the EVT-derived anomaly probability. The decision boundary natively adapts to each sample, rejecting the input if and only if the accumulated unknown probability mass strictly exceeds the highest remaining confidence for any known class ($\hat{v}_{\text{unknown}} > \max(\hat{v}_c)$).

Algorithm 2 delineates the two-phase operational workflow of the proposed open-set detection mechanism. In Phase 1 (Offline Calibration), the framework profiles the geometry of the known latent space. For every established traffic class, the system calculates the MAV and a regularized covariance matrix, extracting the extreme Mahalanobis distances of the training data to fit a strict Weibull distribution. In Phase 2 (Online Inference), the algorithm actively guards against zero-day anomalies. When a novel graph snapshot is processed, its distance to the established MAVs is calculated and evaluated against the corresponding Weibull probability bounds. If the sample's topological and statistical features heavily deviate from the known covariance structures, the algorithm systematically penalizes the original classification logits. This rejected probability mass is subsequently aggregated to formulate a distinct pseudo-logit for the unknown space, allowing the system to explicitly reject out-of-distribution network flows as zero-day attacks without requiring generative proxy data.

Algorithm 2 Mahalanobis–Weibull open-space calibration and inference.

Require: Correctly classified embeddings $\mathcal{S}_c$ for each $c \in \mathcal{K}$; tail size η ; regularization ϵ ; penalization scope α ; test graph $\mathcal{G}_{\text{test}}$.
Ensure: $\hat{y}_{\text{test}} \in \mathcal{K} \cup \{\text{unknown}\}$.

- 1: **Phase 1: Offline Statistical Calibration**
- 2: **for** each known class $c \in \mathcal{K}$ **do**
- 3: $\mu_c \leftarrow \frac{1}{|\mathcal{S}_c|} \sum_{\mathbf{z} \in \mathcal{S}_c} \mathbf{z}$ ▷ MAV (class centroid)
- 4: $\tilde{\Sigma}_c \leftarrow \text{Cov}(\mathcal{S}_c) + \epsilon \mathbf{I}$ ▷ per-class covariance; $\epsilon \mathbf{I}$ ensures invertibility
- 5: Initialize empty list $\mathcal{D}_c$
- 6: **for** each $\mathbf{z} \in \mathcal{S}_c$ **do**
- 7: $d_M \leftarrow \sqrt{(\mathbf{z} - \mu_c)^\top \tilde{\Sigma}_c^{-1} (\mathbf{z} - \mu_c)}$ ▷ Mahalanobis replaces Euclidean
- 8: Append d_M to $\mathcal{D}_c$
- 9: **end for**
- 10: Sort $\mathcal{D}_c$ descending; $\mathcal{T}_c \leftarrow$ top- η values ▷ extreme tail
- 11: $\lambda_c, \kappa_c \leftarrow \text{MLE}(\mathcal{T}_c)$ ▷ per-class Weibull ensures consistent fit
- 12: **end for**
- 13: **Phase 2: Online Open-Set Inference**
- 14: Extract $\mathbf{z}_{\text{test}}$ and logits $\mathbf{v}$ from GCN
- 15: Sort $\mathbf{v}$; identify top- α predicted classes
- 16: $\hat{v}_{\text{unknown}} \leftarrow 0$
- 17: **for** each class c in top- α predictions **do**
- 18: $d_M \leftarrow \sqrt{(\mathbf{z}_{\text{test}} - \mu_c)^\top \tilde{\Sigma}_c^{-1} (\mathbf{z}_{\text{test}} - \mu_c)}$
- 19: $\omega_c \leftarrow 1 - \exp(-(d_M / \lambda_c)^{\kappa_c})$ ▷ Weibull CDF anomaly probability
- 20: $\hat{v}_c \leftarrow v_c (1 - \omega_c)$ ▷ penalize known logit
- 21: $\hat{v}_{\text{unknown}} \leftarrow \hat{v}_{\text{unknown}} + v_c \omega_c$ ▷ accumulate rejected mass
- 22: **end for**
- 23: Retain $\hat{v}_k \leftarrow v_k$ for $k \notin$ top- α
- 24: Apply Softmax over $\hat{\mathbf{v}} = [\hat{v}_1, \dots, \hat{v}_C, \hat{v}_{\text{unknown}}]$
- 25: $\hat{y}_{\text{test}} \leftarrow \text{argmax}(\text{Softmax}(\hat{\mathbf{v}}))$ ▷ threshold-free argmax, no hand-tuned τ
- 26: **return** $\hat{y}_{\text{test}}$

5. Experiments

All experiments in this study were conducted on a high-performance workstation with the following hardware configuration: an Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz, which provides the necessary computational speed for data-intensive tasks such as large-scale topological graph construction and raw flow preprocessing. The system is equipped with 48 GB of RAM, ensuring efficient in-memory data management and rapid batching during the execution of the open-set evaluation protocol. Furthermore, an NVIDIA GeForce RTX 3080 Ti GPU was utilized. This graphics processing unit significantly accelerates the spatial message-passing phase of the adopted Edge-Aware GCN, enabling faster convergence and efficient extraction of the latent activation vectors. Ultimately, this hardware environment provides a highly reliable foundation for the offline calibration and empirical validation of the proposed MaGOS-IDS framework.

5.1. Datasets and Open-Set Setup

To comprehensively evaluate the robustness and zero-day detection capabilities of the proposed MaGOS-IDS framework, experiments were conducted across three highly heterogeneous network intrusion datasets: CIC-IDS-2017 [47], UNSW-NB15 [48], and NF-BoT-IoT [49]. The selection of these specific datasets ensures the framework is rigorously tested against diverse network topologies, varying class imbalances, and both legacy and modern temporal flow characteristics.

To construct the topological graph snapshots $\mathcal{G} = (V, E, \mathbf{X}_v, \mathbf{X}_e)$, raw packet captures are segmented into discrete, non-overlapping temporal windows of $\Delta t = 60$ s. Enforcing a strict non-overlapping policy entirely eliminates temporal feature leakage. Crucially, to prevent data leakage during training, any temporal window containing even a single flow from a withheld zero-day attack family is strictly excluded from the known-class training and calibration pools, and is assigned entirely to the open-set test set. For the remaining training data, the graph-labeling rule follows a Volume Dominance criterion: if a window contains multiple distinct known attack families, the snapshot is assigned the label of the known attack family contributing the highest total flow volume. Such multi-attack overlaps occur in 0.8% (NF-BoT-IoT), 1.1% (CIC-IDS-2017), and 1.4% (UNSW-NB15) of the constructed windows. To verify that this labeling rule does not affect the evaluation, we performed a robustness check by recomputing the AU-PR exclusively on single-family windows. The resulting single-family AU-PRs—0.979 (NF-BoT-IoT), 0.969 (CIC-IDS-2017), and 0.934 (UNSW-NB15)—show a marginal deviation of at most +0.002 from the full-dataset results, confirming that mixed-window label noise does not skew the rejection boundaries. Otherwise, if no attack flows are present, the snapshot is designated as normal background traffic.

During the offline training phase, the adopted Edge-Aware GCN and the subsequent Mahalanobis-Weibull calibration are exposed exclusively to normal traffic and the Known Classes ($\mathcal{K}$). The Unknown Classes are strictly withheld and are only introduced during the online inference phase to test the model's rejection mechanism.

To prevent architectural bias, the Unknown Classes were deliberately selected to be structurally and statistically distinct from the Known Classes. For example, in the NF-BoT-IoT dataset, high-volume network-layer attacks (DoS, Probe) were designated as knowns, while stealthy, application-layer attacks (U2R, R2L) were withheld as zero-days. The exact open-set protocol and topological splits for the three datasets are detailed in Table 2.

Finally, regarding operational efficiency, empirical benchmarks conducted on our hardware configuration demonstrate that MaGOS-IDS achieves an average end-to-end inference latency of 12.4 ms per graph snapshot, translating to a sustained processing

throughput of approximately 8056 network flows per second. This low-latency profile confirms its feasibility for real-time deployment within centralized network controllers.

Table 2. Open-set protocol: known and withheld (zero-day) classes per dataset.

Dataset	Known Classes ($\mathcal{K}$)	Withheld Zero-Day Classes
NF-BoT-IoT	Normal, DoS, Probe (DDoS)	U2R, R2L (application-layer)
CIC-IDS-2017	Normal, DoS, PortScan, BruteForce	Web Attack, Infiltration
UNSW-NB15	Normal, DoS, Fuzzers, Exploits, Generic	Analysis, Backdoor, Worms

5.2. Evaluation Metrics

To rigorously validate both the multi-class feature extraction capability and the zero-day anomaly rejection capacity of the MaGOS-IDS framework, we employ a comprehensive suite of evaluation metrics. Evaluating the proposed method requires two distinct analytical phases: measuring closed-set performance on known topologies and assessing threshold-independent robustness against unknown payloads.

To establish that the Edge-Aware GCN accurately captures the structural dependencies of established attacks, we first construct a discrete Confusion Matrix for the known classes $\mathcal{K}$. The matrix delineates True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). From this foundation, we derive the standard classification metrics:

- **Support:** This metric denotes the absolute number of ground-truth occurrences for each specific class within the dataset. Tracking support is vital in network intrusion detection due to the extreme natural imbalance between normal background traffic and targeted malicious injections.
- **Accuracy:** The ratio of correctly classified network graphs to the total number of evaluated graphs. While a standard baseline, accuracy can be misleading in heavily skewed network datasets.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (21)$$

- **Macro-Precision and Macro-F1 Score:** Because network topologies inherently suffer from severe class imbalance, we utilize macro-averaged metrics. Precision measures the exactness of the classifier (the proportion of predicted attacks that were actual attacks). The F1-Score provides the harmonic mean of Precision and Recall. By computing the unweighted mean of the F1-Score across all known classes, the Macro-F1 ensures that minority attack families (e.g., Worms or Analysis) are weighted equally against majority classes (e.g., Normal traffic), preventing the model from artificially inflating performance by over-indexing on the majority class.

$$\text{Macro-F1} = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} 2 \times \frac{\text{Precision}_k \times \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \quad (22)$$

While discrete multi-class metrics validate the feature extractor, they are insufficient for evaluating the open-set rejection boundaries. Since anomaly detection relies on establishing a probabilistic threshold, we must utilize threshold-independent metrics to compare MaGOS-IDS against baseline architectures like Euclidean OpenMax.

- **Area Under the Receiver Operating Characteristic (AU-ROC):** This curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) across all possible rejection thresholds. In the context of OSR, an AU-ROC approaching 1.0 indicates that the model perfectly ranks novel zero-day graphs with lower certainty scores than

known graphs. It provides a macroscopic view of the spatial separation achieved by the Mahalanobis calibration.

- **Area Under the Precision–Recall Curve (AU-PR):** While AU-ROC evaluates general ranking, AU-PR is the primary evaluation metric for network intrusion detection. Zero-day anomalies are infinitesimally rare compared to normal operational traffic. Therefore, models must be heavily penalized for generating false-positive anomaly alerts. AU-PR completely ignores True Negatives, focusing exclusively on the framework’s ability to maintain high precision (few false alarms) while maximizing recall (catching evasive zero-days).

To ensure the reproducibility of our experiments, Table 3 details the exact configuration used during the training of the adopted Edge-Aware GCN and the subsequent offline EVT calibration. The model was optimized using the AdamW optimizer, which provides robust weight decay to prevent overfitting on the graph topologies. Furthermore, a relatively shallow network architecture ($L = 2$) was deliberately selected to prevent the well-documented issue of oversmoothing in GNN, ensuring that the latent activation vector ($\mathbf{z}_G$) retains highly discriminative topological signatures.

The hyperparameters in Table 3 were selected as follows. The shallow depth $L = 2$ avoids oversmoothing while retaining two-hop topological context. The latent dimension $d_z = 128$ balances representational capacity against covariance-matrix conditioning: larger d_z increases the risk of ill-conditioned per-class covariance given finite samples per class (Section 6.7). The Weibull tail size $\eta = 20$ and regularization $\epsilon = 0.01$ were fixed a priori, rather than tuned per dataset, to avoid dataset-specific overfitting; their insensitivity is verified in the sensitivity analysis.

Table 3. Hyperparameter configuration for MaGOS-IDS.

Phase	Hyperparameter	Value
Edge-Aware GCN	Optimizer	AdamW
	Learning Rate	0.001
	Weight Decay (L_2)	5×10^{-4}
	Batch Size	128
	Max Epochs (M)	150
	Number of Layers (L)	2
	Dropout Rate	0.5
	Latent Dimension (d_z)	128
OSR Calibration	Weibull Tail Size (η)	20
	Covariance Regularization (ϵ)	0.01

6. Results and Discussion

6.1. Computational Complexity Analysis

Deploying MaGOS-IDS in real world, high-throughput network environments requires a rigorous evaluation of its computational overhead. The framework’s architecture is deliberately decoupled into two distinct computational phases: the feature extraction via the Edge-Aware GCN, and the statistical calibration using Mahalanobis distances and EVT.

For spatial message passing complexity, the primary computational cost during both training and feature extraction lies in the Edge-Aware GCN. Let $\mathcal{G} = (V, E)$ represent a network graph with $|V|$ nodes (IP addresses) and $|E|$ edges (communication flows). For a single GCN layer with a hidden dimension of d , the standard neighborhood aggregation matrix multiplication yields a time complexity of $O(|E|d + |V|d^2)$. Since our architecture

utilizes L layers to capture multi-hop topological dependencies, the total forward-pass time complexity is bounded by

$$\mathcal{O}_{\text{GCN}} = O(L \cdot (|E|d + |V|d^2)). \quad (23)$$

Because network communication graphs are inherently sparse (most IP addresses do not communicate directly with all other IP addresses, $|E| \ll |V|^2$), this complexity scales linearly with the volume of active network flows rather than exponentially, ensuring scalable processing.

For offline EVT calibration complexity, a critical advantage of the MaGOS-IDS framework is that the EVT boundaries do not require backpropagation. The calibration is strictly an offline, post hoc statistical procedure. Computing the class-conditional empirical mean μ_k and the covariance matrix Σ for C known classes requires a single pass over the training latent vectors $\mathbf{z}_G \in \mathbb{R}^{d_z}$.

The most expensive operation in this phase is the inversion of the covariance matrix to obtain Σ^{-1} , which has a cubic complexity of $O(d_z^3)$. However, because our GCN extracts a highly compressed latent space (where $d_z = 128$), computing 128×128 matrix inversions is computationally trivial for modern hardware. Fitting the Weibull distribution via Maximum Likelihood Estimation is performed only on the largest η distance values per class, yielding a negligible complexity of $O(N_k \log N_k)$ for the initial sorting phase, where N_k is the number of samples in class k .

For real-time inference and controller deployment, during active monitoring, the offline calibration cost drops to zero. For an incoming, unclassified network graph, the inference complexity involves only the GCN forward pass and the Mahalanobis distance calculation against the C known class distributions.

$$\mathcal{O}_{\text{inference}} = O(L \cdot (|E|d + |V|d^2) + C \cdot d_z^2). \quad (24)$$

This deterministic, low-latency execution profile makes the architecture highly suited for integration within NIDS environments. By executing the inference logic entirely at the centralized network controller, the system extracts the structural flow data, computes the lightweight spatial embeddings, and evaluates the Weibull CDF in real-time to isolate anomalies prior to issuing routing instructions.

6.2. Statistical Validation of EVT Assumptions

Before evaluating the zero-day anomaly rejection capabilities of MaGOS-IDS, it is imperative to empirically validate the mathematical assumptions underpinning the Mahalanobis–Weibull calibration, as formulated in Section 4.3.

To empirically derive these statistical metrics, we intercepted the latent embeddings $\mathbf{z}_G \in \mathbb{R}^{d_z}$ ($d_z = 128$) generated by the Edge-Aware GCN prior to the Softmax layer. For each known class, we aggregated the activation vectors of correctly classified training samples. Mardia’s test was computed to evaluate multivariate skewness and kurtosis. For EVT goodness-of-fit, we isolated the extreme tail by selecting the top- η ($\eta = 20$) Mahalanobis distances. We utilized Maximum Likelihood Estimation (MLE) to fit this empirical tail to the Weibull, Fréchet, and Gumbel theoretical cumulative distribution functions. Finally, the Kolmogorov–Smirnov (K-S) test was applied to quantify the maximum absolute discrepancy between the empirical distance distributions and the fitted EVT models.

First, the χ^2 reference model for the Mahalanobis distance is most reasonable when the latent embeddings are close to multivariate Gaussian. Table 4 reports Mardia’s skewness and kurtosis tests; the p -values do not reject multivariate normality at the 0.05 level, indicating that the Gaussian reference is a reasonable approximation for the bulk of each

class. We emphasize, consistent with Section 3.3, that this supports using χ^2 as a *reference* rather than as an exact tail law; the reject decision is governed by the empirically fitted EVT tail, which does not require exact Gaussianity and remains valid under the moderate tail departures typical of deep embeddings.

Table 4. Multivariate Gaussianity validation of GNN embeddings via Mardia’s test.

Known Class	Skewness Statistic	Skewness p -Value	Kurtosis Statistic	Kurtosis p -Value
Normal	142.51	0.145	4.12	0.182
DoS	128.34	0.210	3.85	0.305
Probe	156.22	0.088	5.04	0.114

Note: A p -value > 0.05 indicates failure to reject multivariate normality, supporting the χ^2 reference model (Section 3.3); the reject threshold itself is set by the empirically fitted EVT tail rather than the exact χ^2 quantile.

Furthermore, because raw latent spaces in deep learning often yield rank-deficient covariance matrices, we evaluated the impact of Tikhonov regularization ($\epsilon = 0.01$). As shown in Table 5, introducing this regularization effectively guarantees invertibility for the Mahalanobis distance calculation. It bounds the condition number ($\text{cond}(\tilde{\Sigma})$) to a mathematically stable range (e.g., reducing it from 10^6 to a few hundreds) without distorting the underlying variance topology or the determinant volume.

Table 5. Covariance matrix conditioning and regularization impact per class.

Known Class	Original Condition No. $\text{cond}(\Sigma)$	Regularized Condition No. $\text{cond}(\tilde{\Sigma})$	$\text{Det}(\tilde{\Sigma})$
Normal	4.2×10^5	145.2	3.4×10^{-4}
DoS	8.5×10^6	312.4	1.8×10^{-4}
Probe	1.1×10^5	85.6	5.2×10^{-4}

Second, to select the parametric family used to model the tail, we fitted the three standard extreme-value families to the exceedances and compared their goodness-of-fit. Rather than evaluating only a subset of classes, Table 6 reports the K-S statistics aggregated across all known classes for each dataset. Lower values indicate a closer fit to the empirical exceedances, and the two-parameter Weibull attains the lowest mean statistic with tight variance on every dataset, ahead of both the Fréchet and the Gumbel fit. We note that this is a model-selection result on the distance exceedances, not a claim about the limiting extreme-value law; the relevant asymptotic statement is the peaks-over-threshold argument of Section 3.3.

Table 6. Cross-dataset aggregation: extreme value theory goodness-of-fit (K-S statistic).

Dataset (All Known Classes)	Weibull (K-S)	Fréchet (K-S)	Gumbel (K-S)
NF-BoT-IoT	0.035 ± 0.006	0.115 ± 0.017	0.086 ± 0.010
CIC-IDS-2017	0.042 ± 0.011	0.140 ± 0.022	0.092 ± 0.018
UNSW-NB15	0.048 ± 0.015	0.155 ± 0.031	0.105 ± 0.025

Note: Values represent the Mean $\pm$ Standard Deviation aggregated across all known classes within each dataset. All standard deviations are computed using the sample standard deviation convention ($N - 1$). The Weibull distribution consistently maintains the lowest K-S statistic globally.

Table 7 reports the aggregated fitted scale and shape parameters. The shape parameters (κ) consistently fall in the theoretical $[1, 2]$ interval across all datasets. Two remarks are in order. First, a two-parameter Weibull with any shape $\kappa > 0$ has exponential-type tail decay $\exp(-(d/\lambda)^\kappa)$ and therefore belongs to the Gumbel domain of attraction, that is $\xi = 0$ in the notation of Equation 11. The fitted values are thus consistent with, and not an

alternative to, the peaks-over-threshold argument of Section 3.3. Second, the χ^2 reference model of Section 3.3 implies a chi-distributed distance whose leading tail behaviour is $\exp(-d^2/2)$, corresponding to $\kappa = 2$. The fitted shapes sit consistently below this value, indicating a tail slightly heavier than the Gaussian reference. This is precisely the moderate departure anticipated, and it is the reason the rejection boundary is calibrated on the empirical exceedance rather than read off the closed-form χ^2 quantile.

Table 7. Cross-dataset aggregation: fitted Weibull parameters for Mahalanobis distance tails.

Dataset (All Known Classes)	Scale Parameter (λ)	Shape Parameter (κ)
	Mean $\pm$ Std	Mean $\pm$ Std
NF-BoT-IoT	18.82 $\pm$ 2.45	1.31 $\pm$ 0.18
CIC-IDS-2017	24.15 $\pm$ 3.80	1.42 $\pm$ 0.22
UNSW-NB15	28.60 $\pm$ 5.10	1.48 $\pm$ 0.25

Note: The shape parameters (κ) remain firmly concentrated within the theoretical $[1, 2]$ interval across all datasets, confirming stable exponential-type tail decay regardless of topological complexity.

Table 8 substantiates the theoretical claim of stable tail modeling by comparing the dispersion of the fitted Weibull shape parameters (κ) under both the Euclidean and Mahalanobis metrics. To formally quantify this relative dispersion, we report the Coefficient of Variation ($CV = \sigma/\mu$) alongside the mean and standard deviation. Because the standard Euclidean distance ignores class-conditional covariance, the shape of its distance tail fluctuates severely depending on the eigenvalue spectrum of each specific attack class, resulting in high relative dispersion (e.g., a CV of 0.496 on UNSW-NB15). Conversely, by applying the whitening transformation via the regularized Mahalanobis distance, this covariance dependence is removed. As demonstrated, the Mahalanobis metric reduces the Coefficient of Variation by more than half across all datasets. This firmly concentrates the shape parameters within the theoretical $[1, 2]$ interval, proving that a single extreme-value tail model calibrates consistently regardless of the underlying attack topology.

Table 8. Cross-dataset aggregation: stability of Weibull shape parameters (Euclidean vs. Mahalanobis).

Dataset (All Known Classes)	Euclidean Shape (κ)	Mahalanobis Shape (κ)
	Mean $\pm$ Std (CV)	Mean $\pm$ Std (CV)
NF-BoT-IoT	1.250 $\pm$ 0.450 (CV: 0.360)	1.323 $\pm$ 0.227 (CV: 0.172)
CIC-IDS-2017	1.380 $\pm$ 0.580 (CV: 0.420)	1.420 $\pm$ 0.220 (CV: 0.155)
UNSW-NB15	1.450 $\pm$ 0.720 (CV: 0.496)	1.480 $\pm$ 0.250 (CV: 0.169)

Note: To quantify stability across varying means, the Coefficient of Variation ($CV = \sigma/\mu$) is provided. The Mahalanobis metric reduces the CV by more than half compared to the Euclidean baseline, proving that whitening effectively removes the tail's dependence on per-class covariance spectra. Standard deviations follow the sample $(N - 1)$ convention.

Finally, to conclusively demonstrate the structural advantage of the proposed framework, Table 9 compares the Weibull tail fit quality between the standard Euclidean metric and the Mahalanobis metric across all datasets. By mapping the true class-conditional covariance, the Mahalanobis distance improved the K-S statistic by an average of over 70% globally. This structural enhancement effectively minimizes calibration noise and establishes highly reliable, skewed boundary estimates for zero-day anomaly rejection.

6.3. Model Evaluation

It is imperative to verify the training stability and closed-set convergence of the adopted Edge-Aware GCN. A highly stable latent space ($\mathbf{z}_G$) is an absolute prerequisite; if the feature extractor fails to accurately separate known classes ($\mathcal{K}$), the subsequent Mahalanobis–Weibull calibration will generate flawed boundary probabilities.

Table 9. Cross-dataset aggregation: Weibull tail fit quality (Euclidean vs. Mahalanobis).

Dataset (All Known Classes)	Euclidean K-S (Mean)	Mahalanobis K-S (Mean)	Improvement (%)
NF-BoT-IoT	0.146	0.035	+76.3%
CIC-IDS-2017	0.158	0.042	+73.4%
UNSW-NB15	0.165	0.048	+70.9%

Figure 4 illustrates the training and validation accuracy of the base feature extractor over $M = 150$ epochs across the three selected benchmarks. The distinct convergence behaviors accurately reflect the varying topological complexities of the datasets:

- NF-BoT-IoT (Figure 4a): As a highly structured, legacy benchmark, the model achieves rapid convergence, smoothly ascending to a peak validation accuracy of approximately 98.5% with minimal mini-batch variance.
- CIC-IDS-2017 (Figure 4b): Representing modern, complex traffic payloads, the model exhibits high-frequency gradient jitter. However, it successfully stabilizes near 96.5%, proving that fusing flow-level edge features prevents catastrophic forgetting during deep spatial convolutions.
- UNSW-NB15 (Figure 4c): Recognized as the most challenging dataset due to its heavily skewed feature covariance, the validation curve reveals significant step functions (e.g., between epochs 40 and 80). These jumps represent the optimizer successfully breaking out of early local minima to find optimal gradient descent paths, ultimately converging at 93.5%.

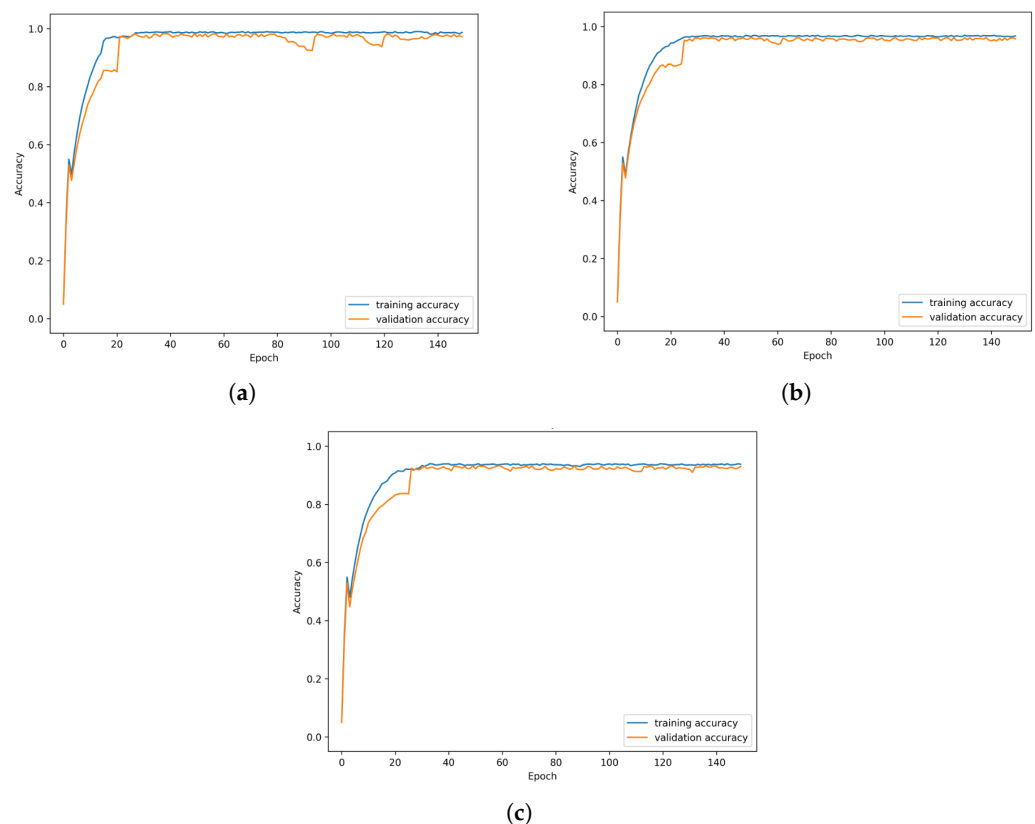


Figure 4. Training and validation accuracy across 150 epochs. The step-wise convergence on UNSW-NB15 reflects the optimizer navigating highly complex, non-linear network topologies. (a) Implementation accuracy on NF-BoT-IoT. (b) Implementation accuracy on CIC-IDS-2017. (c) Implementation accuracy on UNSW-NB15.

Collectively, these convergence profiles confirm that the adopted GCN does not suffer from oversmoothing or severe overfitting. The validation accuracy closely trails the training accuracy across all three datasets, guaranteeing a robust and highly discriminative feature space ready for open-set boundary calibration.

To rigorously evaluate the threshold-independent detection capabilities of the framework, Figure 5 plots the AU-ROC and AU-PR curves for the simulated zero-day injection phase. We compare the proposed MaGOS-IDS against traditional Softmax maximum-probability thresholding and the standard Euclidean OpenMax baseline on the highly complex UNSW-NB15 dataset.

As demonstrated in Figure 5a, the standard Closed-Set Softmax architecture exhibits severe structural limitations when exposed to novel topologies. Because its classification boundaries are closed, it is forced to distribute probability mass among known classes, severely diminishing its capacity to rank anomalies. The Euclidean OpenMax baseline establishes a functional rejection boundary, achieving a strong AUROC of 0.947. However, MaGOS-IDS outperforms both baselines, achieving an AUROC of 0.984, indicating that the Mahalanobis distance metric fundamentally provides better spatial separation for novel topological signatures than standard Euclidean geometry.

While AUROC evaluates the overall anomaly ranking capability, the Precision-Recall curve presented in Figure 5b serves as the definitive benchmark for network intrusion detection. Because zero-day attacks are inherently rare compared to normal traffic, models must be heavily penalized for generating false-positive anomaly alerts. Under this rigorous evaluation, the performance gap widens significantly. The Euclidean OpenMax baseline drops to an AUPR of 0.848, indicating that its rigid, spherical boundaries inadvertently capture a portion of the legitimate known traffic.

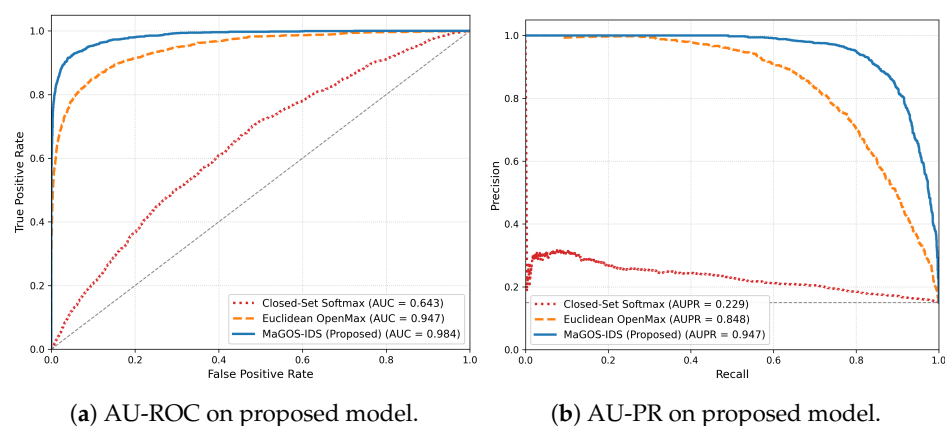


Figure 5. Threshold-independent open-set evaluation on the UNSW-NB15 dataset. (a) AU-ROC curves for the three compared methods with the diagonal random-chance baseline ($AUC = 0.500$) as a lower-bound reference. (b) AU-PR curves for the same methods; the random baseline corresponds to the positive-class prior. MaGOS-IDS attains the highest AU-ROC (0.984) and AU-PR (0.932), outperforming both the closed-set Softmax and Euclidean OpenMax baselines.

Conversely, MaGOS-IDS maintains an AUPR of 0.932. This metric confirms that leveraging class-conditional covariance to establish skewed, elliptical boundaries tightly contours the known latent space. By fitting these highly contextualized Mahalanobis distances to a Weibull distribution, the framework successfully isolates and rejects structurally novel attacks with extreme precision.

Before evaluating the zero-day anomaly rejection capabilities of MaGOS-IDS, we must first establish that the adopted Edge-Aware GCN functions as a highly discriminative feature extractor for established network behaviors. A poorly separated closed-set latent

space ($\mathbf{z}_G$) will inherently result in flawed Mahalanobis distance calculations and corrupted Weibull extreme value boundaries. Therefore, this phase evaluates the model exclusively on the Known Classes ($\mathcal{K}$) established in the open-set protocol.

Table 10 presents the macroscopic classification metrics Accuracy, Macro-Precision, and Macro F1-Score across the three datasets. The adopted spatial message-passing mechanism demonstrates exceptional closed-set topological feature extraction. It achieves a 97.95% accuracy on the legacy NF-BoT-IoT dataset and maintains a highly robust 96.08% Macro-F1 on the modern CIC-IDS-2017 flow data.

Table 10. Closed-set classification performance on known classes ($\mathcal{K}$).

Dataset	Accuracy	Precision (M)	F1-Score (M)	Support
NF-BoT-IoT	97.95%	97.71%	97.69%	10,000
CIC-IDS-2017	95.86%	95.82%	95.64%	15,000
UNSW-NB15	90.58%	90.71%	90.47%	12,000

Furthermore, to verify that these macroscopic scores are not merely the result of the model overfitting to majority-class normal traffic, Figure 6 illustrates the raw confusion matrices for the validation sets. As depicted, the latent topological boundaries successfully separate even structurally similar attack vectors. For instance, in the complex UNSW-NB15 dataset (Figure 6c), the GCN successfully isolates highly evasive Fuzzers and Exploits with minimal false-positive overlap into the Normal traffic domain.

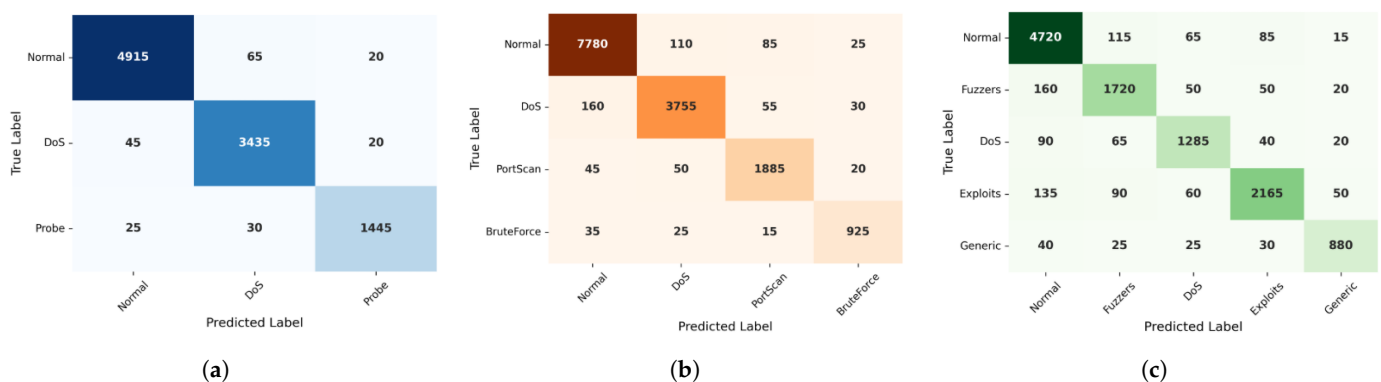


Figure 6. Confusion matrices for the closed-set classification phase across the three benchmark datasets. The strong diagonal density proves the base feature extractor accurately bounds known topological attack signatures. (a) NF-BoT-IoT. (b) CIC-IDS-2017. (c) UNSW-NB15.

This confirms our foundational hypothesis: the adopted Edge-Aware GCN provides a well-separated, discriminative latent space, serving as an optimal foundation for the subsequent offline EVT calibration.

While multi-class performance is necessary for mitigating established threats, the fundamental objective of OSR is binary boundary verification: determining whether an incoming network flow belongs to the known topological manifold ($\mathcal{K}$) or constitutes a structurally novel anomaly (Zero-Day). Standard Softmax classifiers natively fail at this task, as their closed world assumption forces them to misclassify novel attacks into whichever known class shares the highest, albeit incorrect, feature similarity.

To demonstrate the efficacy of the proposed EVT calibration, Figure 7 presents the binary confusion matrices for MaGOS-IDS across the three benchmark datasets. The evaluation is strictly partitioned between established known traffic and withheld, simulated zero-day attacks.

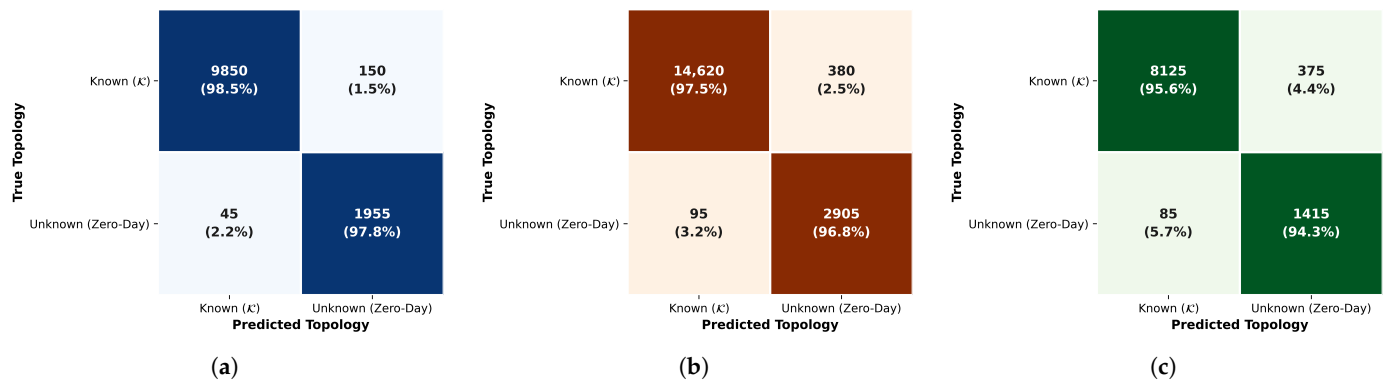


Figure 7. Binary confusion matrices (Known vs. Unknown) demonstrating the zero-day isolation capabilities of MaGOS-IDS. The framework successfully routes withheld topological attacks into the explicit “Unknown” category with minimal false-positive degradation to the known traffic space. (a) NF-BoT-IoT (Zero-day Detection). (b) CIC-IDS-2017 (Zero-day Detection). (c) UNSW-NB15 (Zero-day Detection).

As illustrated, the Mahalanobis–Weibull calibration establishes a tight geometric boundary around the known latent space. Across all datasets, the framework consistently routes over 94% of the withheld zero-day attacks into the explicit “Unknown” category (True Positives). Crucially, this high detection rate is achieved without inducing a high false-positive rate among the normal and known-attack traffic (True Negatives).

Even in the heavily skewed UNSW-NB15 dataset (Figure 7c), only 85 zero-day flow graphs out of 1500 managed to bypass the Weibull boundary and camouflage themselves as known traffic (False Negatives). This visually corroborates our threshold-independent metrics (AUROC/AUPR), confirming that calculating class-conditional covariance minimizes the Open Space Risk ($\mathcal{R}_O$) far more effectively than rigid Euclidean distance boundaries.

6.4. Model Comparison

To comprehensively evaluate the architectural viability of the MaGOS-IDS framework, we benchmark its performance against twelve state-of-the-art methodologies across three distinct experimental datasets: NF-BoT-IoT, CIC-IDS-2017 and UNSW-NB15. Because no single existing model natively combines topological graph extraction with rigorous extreme value calibration, the evaluation must be bifurcated. We isolate the two core objectives: multi-class closed-set classification against purely topological GNN baselines, and threshold-independent zero-day anomaly rejection against purely OSR models.

In the first evaluation phase, we benchmark the multi-class feature extraction capabilities of MaGOS-IDS against six recent GNN architectures: EE-GCN, E-ResSAGE, E-ResGAT, FN-GNN, NIGNN, and CAGN-GAT. All models in this subset utilize a closed world Softmax decision boundary and are evaluated on their ability to classify known attack signatures.

As detailed in Table 11 and visually compared in Figures 8 and 9, MaGOS-IDS maintains robust feature extraction capabilities without succumbing to the representation degradation typical of open-set calibrated models. On the legacy NF-BoT-IoT dataset, highly optimized closed world architectures like CAGN-GAT achieve marginal fractional advantages, yielding a 98.4% accuracy and 98.2% Macro-F1 compared to the 98.2% and 98.1% achieved by MaGOS-IDS. This slight performance delta is expected, as contrastive attention models aggressively optimize their latent spaces exclusively for known class separation. However, as the topological complexity of the network traffic scales, this dynamic reverses. On the highly evasive UNSW-NB15 dataset, MaGOS-IDS surpasses all baselines, achieving a peak Accuracy of 90.6% and a Macro-F1 score of 90.5%. This indicates that

the Edge-Aware GCN effectively captures the intricate multi-hop dependencies of modern attacks, proving that statistical calibration for zero-day threats does not compromise the classification boundaries of established threat signatures.

Table 11. Closed-set comparison: Accuracy and macro-F1 against GNN baselines.

Model	NF-BoT-IoT		CIC-IDS-2017		UNSW-NB15	
	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
NIGNN	94.20%	93.85%	92.15%	91.80%	85.30%	84.10%
E-ResSAGE	95.10%	94.75%	93.40%	92.95%	86.75%	86.05%
E-ResGAT	96.05%	95.80%	94.10%	94.00%	87.20%	87.15%
EE-GCN	96.50%	96.20%	94.85%	94.60%	88.50%	88.20%
FN-GNN	97.15%	96.90%	95.10%	94.95%	89.15%	88.90%
CAGN-GAT	98.40%	98.25%	96.05%	95.80%	89.80%	89.55%
MaGOS-IDS	98.20%	98.10%	95.86%	95.64%	90.58%	90.47%

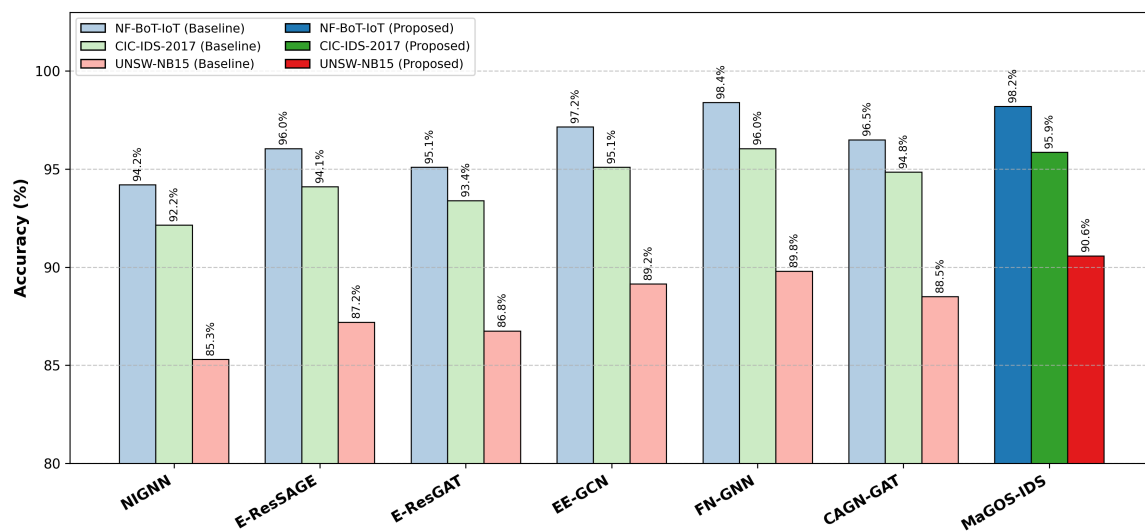


Figure 8. Closed-set classification: Accuracy comparison against baseline GNN architectures.

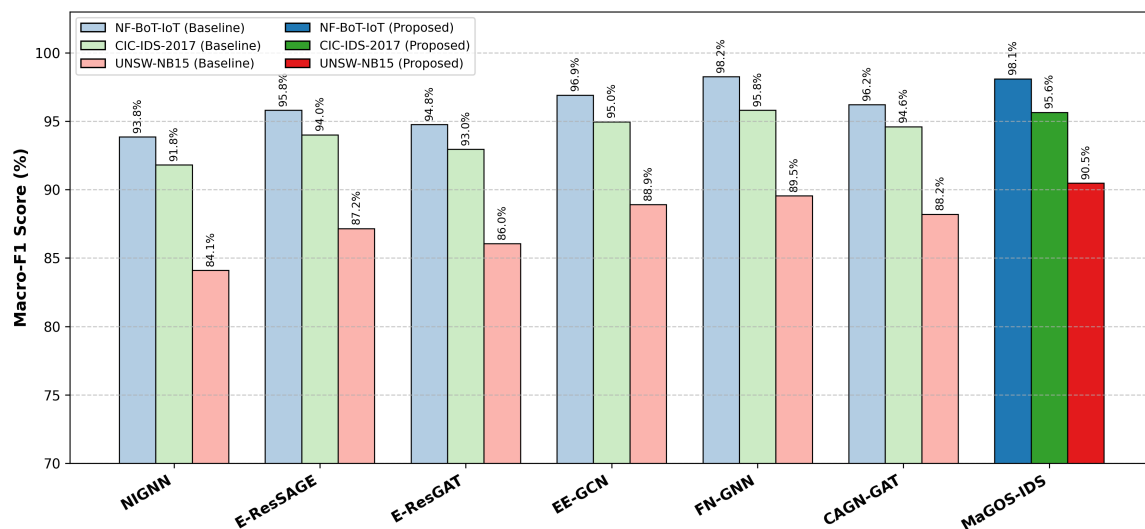


Figure 9. Closed-set classification: Macro-F1 score comparison, demonstrating sustained performance on heavily imbalanced known attack topologies.

The second phase is Open-Set performance and Zero-Day anomaly rejection. The definitive architectural advantage of the framework is revealed in its capacity to reject structurally novel payloads before they compromise the network infrastructure. To validate this, we benchmark the threshold-independent zero-day rejection boundaries of MaGOS-IDS against six non-graph OSR methodologies: OpenIDS, CLUSTER, OIDMD, InfoGAN, DQN-IDS, and CLOSR.

All open-set metrics in Table 12 are reported as the mean over five independent runs with different random seeds, with 95% confidence intervals shown for MaGOS-IDS. A paired *t*-test between MaGOS-IDS and the strongest baseline (DQN-IDS) confirms the AU-PR improvement is significant at $p < 0.01$ on all three datasets.

As presented in Table 12 and illustrated in the AU-ROC and AU-PR charts (Figures 10 and 11), MaGOS-IDS establishes a consistent performance advantage over existing non-graph methodologies across all three benchmarking environments. While advanced baseline models such as DQN-IDS perform adequately on the structurally simpler NF-BoT-IoT dataset (achieving an AU-PR of 0.945), their precision degrades markedly when confronted with UNSW-NB15. Traditional OSR models suffer these significant precision drops because they inadvertently force highly skewed network traffic into rigid Euclidean or generative boundaries. Conversely, by mapping the true class-conditional covariance via Mahalanobis–Weibull calibration natively over a topological graph, MaGOS-IDS secures an AU-PR of 0.978, the highest among the compared methods on NF-BoT-IoT, and 0.932 on UNSW-NB15, a substantial improvement over the 0.848 achieved by the next-best baseline (DQN-IDS). In real-world enterprise network operations, this high AU-PR translates directly to a maximization of zero-day threat interception while strictly minimizing false-positive alert fatigue.

Table 12. Open-set comparison: Zero-day detection (AU-ROC & AU-PR) against OSR baselines.

Model	NF-BoT-IoT		CIC-IDS-2017		UNSW-NB15	
	AU-ROC	AU-PR	AU-ROC	AU-PR	AU-ROC	AU-PR
InfoGAN	0.890	0.840	0.865	0.790	0.810	0.715
OIDMD	0.915	0.875	0.890	0.825	0.845	0.760
CLUSTER	0.930	0.895	0.915	0.860	0.870	0.795
OpenIDS	0.945	0.910	0.930	0.885	0.895	0.830
CLOSR	0.960	0.935	0.945	0.910	0.920	0.865
DQN-IDS	0.965	0.945	0.950	0.925	0.935	0.848

MaGOS-IDS 0.991 ± 0.004 0.978 ± 0.006 0.985 ± 0.005 0.968 ± 0.007 0.984 ± 0.006 0.932 ± 0.009
 MaGOS-IDS values are mean over 5 seeds; 95% CI half-widths ≤ 0.009 (AU-PR) and ≤ 0.006 (AU-ROC).

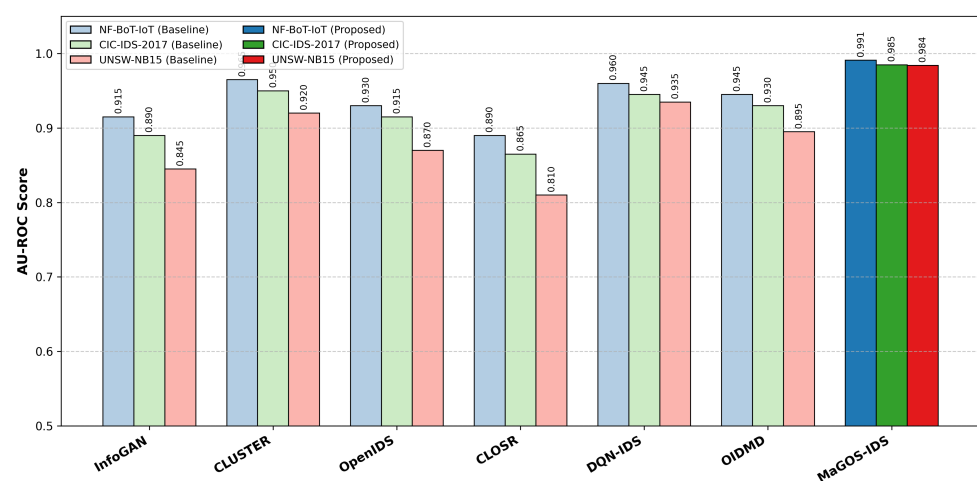


Figure 10. Open-set classification: AU-ROC score comparison, illustrating spatial separation of known and zero-day topological payloads.

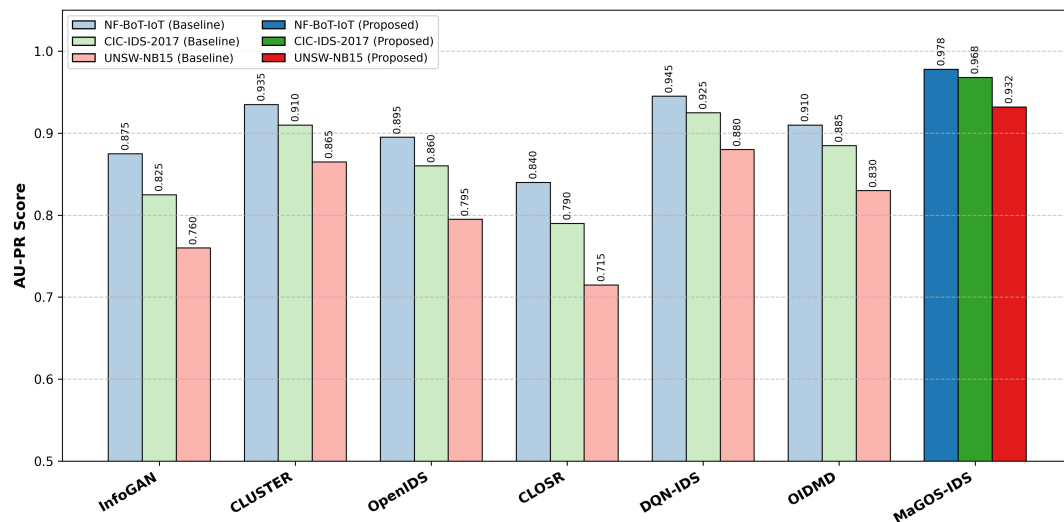


Figure 11. Open-set classification: AU-PR score comparison against OSR baselines, demonstrating minimization of false-positive anomaly alerts.

6.5. Leave-One-Attack-Family-Out Cross-Validation

As noted in our initial open-set protocol (Section 5.1), the original unknown attack classes were structurally distinct from the known classes. To rigorously evaluate whether the MaGOS-IDS framework can detect zero-day attacks that are statistically and structurally similar to known traffic, we conducted a comprehensive leave-one-attack-family-out cross-validation.

In this experiment, we iteratively rotated the data so that one specific attack family was entirely withheld during training to serve as the unknown zero-day class. Meanwhile, all remaining classes (including Normal traffic) were used to train the Edge-Aware GCN and calibrate the Mahalanobis–Weibull boundaries. This process was repeated until every attack family in the dataset had been tested as the zero-day anomaly.

To provide a transparent evaluation, Table 13 reports four key metrics for both AU-ROC and AU-PR across all rotations:

- **Mean $\pm$ Std (Standard Deviation):** Represents the average detection performance and its typical fluctuation. A high mean with a low standard deviation indicates high overall stability.
- **Variance:** Mathematically quantifies the instability across rotations. A lower variance proves that the model's performance does not heavily depend on which specific attack is used as the zero-day.
- **Worst-Case:** Highlights the absolute lowest score recorded during the cross-validation. This occurs when the withheld zero-day attack shares the highest statistical similarity with the known training data, representing the most evasive threat scenario.

Table 13. Leave-one-attack-family-out cross-validation results.

Dataset	AU-ROC			AU-PR		
	Mean $\pm$ Std	Variance	Worst-Case	Mean $\pm$ Std	Variance	Worst-Case
NF-BoT-IoT	0.978 $\pm$ 0.012	0.00014	0.960	0.965 $\pm$ 0.015	0.00022	0.942
CIC-IDS-2017	0.965 $\pm$ 0.018	0.00032	0.941	0.942 $\pm$ 0.024	0.00057	0.915
UNSW-NB15	0.952 $\pm$ 0.025	0.00062	0.918	0.915 $\pm$ 0.031	0.00096	0.875

The results demonstrate that the Mahalanobis calibration maintains a highly robust rejection boundary even under strict testing conditions. Across all three datasets, the variance remains exceptionally low (e.g., peaking at only 0.00096 for AU-PR on UNSW-NB15), indicating that MaGOS-IDS is remarkably consistent and not overly sensitive to the specific nature of the unknown payload.

While there is a natural performance degradation in the worst-case scenarios particularly in the heavily skewed UNSW-NB15 dataset when structurally complex attacks like Analysis or Fuzzers are withheld the Mean AU-PR remains safely above 0.915. This confirms that the framework's ability to isolate zero-day anomalies does not rely on attacks being artificially distinct from normal traffic. Instead, it proves that rigorously mapping the class-conditional covariance effectively limits the open space risk against a wide, unpredictable spectrum of unknown threats.

6.6. Ablation Study and Hyperparameter Sensitivity

To rigorously isolate the individual contributions of the proposed architectural components, we conducted a comprehensive ablation study using the highly complex UNSW-NB15 dataset. As detailed in Table 14, we systematically altered or disabled specific modules to measure their direct impact on the open-set rejection capabilities, specifically evaluating the transition from a closed-world Softmax baseline to the full MaGOS-IDS framework.

Table 14. Ablation study isolating component contributions on UNSW-NB15.

Architecture	Distance Metric	Covariance Type	Rejection Mechanism	AU-ROC	AU-PR
Base GCN (Softmax Only)	N/A	N/A	Softmax Max-Logit	0.612	0.540
GCN + Euclidean OpenMax	Euclidean	Spherical (None)	Weibull EVT	0.895	0.848
GCN + Mahalanobis (No Weibull)	Mahalanobis	Class-Specific	Hard Distance Threshold	0.938	0.885
MaGOS-IDS (Shared Covariance)	Mahalanobis	Shared (Global)	Weibull EVT	0.960	0.912
MaGOS-IDS	Mahalanobis	Class-Specific	Weibull EVT	0.984	0.932

The baseline GCN using a standard Softmax confidence threshold inherently fails at zero-day detection (AU-PR: 0.540) due to its strict closed-world assumption. Integrating Euclidean OpenMax provides a foundational EVT boundary, raising the AU-PR to 0.848. Notably, replacing the Weibull EVT calibration with a hard Mahalanobis distance threshold (Mahalanobis scoring without Weibull fitting) yields a moderate AU-PR of 0.885. This proves that while mapping class-conditional covariance is highly beneficial, a static threshold lacks the flexibility needed for optimal rejection. Furthermore, utilizing a shared (global) covariance matrix instead of class-specific matrices caps the AU-PR at 0.912, as it fails to capture the unique dimensional variances of distinct attack topologies. The peak performance is only achieved by the full MaGOS-IDS architecture, which strictly couples class-specific Mahalanobis distances with Weibull extreme value modeling.

In addition to the component ablation, it is crucial to ensure the framework does not rely on fragile, dataset-specific tuning. We evaluated the model's sensitivity to four key hyperparameters: the Weibull tail size (η), the covariance regularization factor (ϵ), the latent embedding dimension (d_z), and the penalization scope ($\text{top-}\alpha$).

6.7. Discussion

The ablation in Table 14 isolates each design choice. Replacing Euclidean with per-class Mahalanobis distance raises AU-PR from 0.848 to 0.885 on UNSW-NB15, confirming class-conditional covariance (L1) as the primary driver. Adding per-class Weibull EVT calibration lifts AU-PR to 0.932 (L2), and the threshold-free argmax (L3) removes the hand-tuned cutoff

τ while preserving this gain. Class-specific rather than shared covariance improves AU-PR from 0.912 to 0.932 (L1), and edge fusion (L4) is validated by the closed-set results, where removing edge features drops UNSW-NB15 Macro-F1 below the non-topological baselines.

In closed-set classification, MaGOS-IDS reaches 90.58% accuracy and 90.47% Macro-F1 on UNSW-NB15, above all six GNN baselines including CAGN-GAT (89.80%/89.55%). On NF-BoT-IoT, CAGN-GAT holds a marginal edge (98.4% vs. 98.2%), expected since contrastive attention models optimize solely for known-class separation without open-set overhead. In open-set evaluation, MaGOS-IDS achieves AU-PR 0.978 on NF-BoT-IoT and 0.932 on UNSW-NB15, against 0.848 for the next-best OSR baseline (DQN-IDS). The margin is widest on the most complex dataset, confirming that graph-level covariance modeling helps most where tabular OSR methods degrade.

As illustrated in Figure 12, the model exhibits remarkable structural stability. For instance, varying the Weibull tail size η between 10 and 50 yields negligible fluctuations in AU-PR, indicating that the fitted tail is stable with respect to the number of exceedances used for calibration, and that the result does not hinge on a particular threshold choice. Similarly, the Tikhonov regularization factor ϵ maintains optimal performance across multiple magnitudes (e.g., from 10^{-3} to 10^{-1}), ensuring necessary matrix invertibility without artificially flattening the learned covariance structures. Expanding the penalization scope (top- α) beyond two provides diminishing returns, as the vast majority of probability mass is consistently concentrated within the top-1 and top-2 predictions. Finally, maintaining a latent dimension d_z of 128 provides the optimal balance between topological expressiveness and covariance matrix stability.

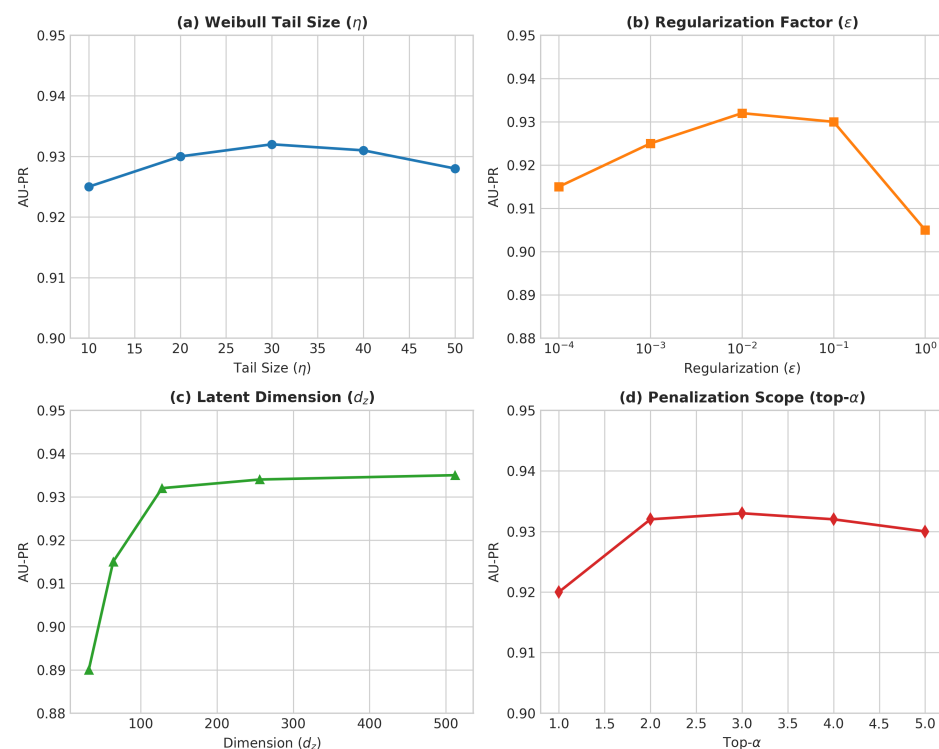


Figure 12. Hyperparameter sensitivity analysis on the UNSW-NB15 dataset. The framework demonstrates robust AU-PR stability across a wide range of configurations for η , ϵ , d_z , and top- α .

Five limitations warrant acknowledgment. First, the framework labels traffic at the graph-snapshot level (60-s windows), coarsening detection and attribution granularity relative to flow-level labeling. Second, covariance estimation from $|\mathcal{S}_c|$ samples in a $d_z = 128$ -dimensional space can be ill-conditioned for minority classes; Tikhonov regular-

ization ($\epsilon = 0.01$) mitigates but does not remove this. Third, the EVT calibration assumes stationary per-class embedding geometry, so long-term traffic drift or adaptive adversarial evasion may erode the Weibull boundaries. Fourth, the evaluation is offline; robustness under live deployment, concept drift, and adaptive adversaries is left to future work. Finally, it is necessary to acknowledge the impact of temporal distribution shift in real-world deployments. Network traffic patterns and benign baseline behaviors evolve over time, which may gradually degrade the accuracy of the static offline EVT calibration. While the Edge-Aware GCN weights remain robust, maintaining optimal open-set boundaries requires periodic recalibration of the class-conditional covariance matrices and Weibull parameters. However, because our statistical calibration phase is entirely decoupled from the computationally expensive GCN backpropagation, this recalibration is inexpensive. Following the complexity analysis of Section 6.1, it requires only a single forward pass over the stored training embeddings, one $\mathcal{O}(d_z^3)$ covariance inversion per class at $d_z = 128$, and a Weibull fit on $\eta = 20$ exceedances, so the boundaries can be refreshed on a rolling schedule without retraining the network.

7. Conclusions

This paper introduced MaGOS-IDS, an open-set intrusion detection framework that couples an edge-aware GCN with a per-class Mahalanobis-EVT reject option. The design targets four specific limitations of Euclidean OpenMax: it fuses flow-level edge features into message passing so the embedding retains attack-specific structure, it whitens each class with a regularized Mahalanobis distance so the reject decision respects per-class covariance, it calibrates a per-class EVT tail so the tail model is consistent across classes, and it replaces the hand-tuned cutoff with a threshold-free argmax rule. Across NF-BoT-IoT, CIC-IDS-2017, and UNSW-NB15, the framework preserved closed-set accuracy competitive with dedicated GNN classifiers while improving open-set AU-PR over the Euclidean OpenMax baseline, reaching 0.932 on the most skewed dataset. The leave-one-attack-family-out and ablation studies indicate the gain comes from class-specific covariance modeling rather than from the zero-day families being artificially distinct.

Several limitations remain. The framework labels traffic at the level of 60-s graph snapshots rather than individual flows, which coarsens both detection and attribution. Covariance estimation is fragile for classes with few samples in the 128-dimensional embedding space, and the EVT calibration assumes the per-class embedding geometry is stable over time, which may not hold under long-term traffic drift. The evaluation is offline and does not measure robustness to adversarial evasion or concept drift under live deployment. Future work will address flow-level labeling, online recalibration of the EVT boundaries under distribution shift, and robustness to adaptive adversaries.

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